

MTD / PhD / EngD / MEng ·

SEPTEMBER – DECEMBER 2026

Innovation by Design (Week 2)

14 WEEKS

4 MILESTONES

5–50 UNITS

1 SALES
GALA/FINAL
PRESENTATION

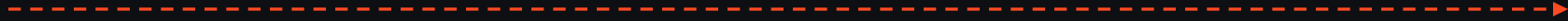
Discover: understanding customer needs; evidence-based customer interviews; observations; AI-assisted desk research and needs synthesis.

Faculty Lead Dr Aba Kang · Co-Leads Raghavan Nagarajan, Huang Shaoying

Main agenda: Course brief and Team formation

PAY ATTENTION

PROJECT TOPIC/CONSTRAINTS (Updated)

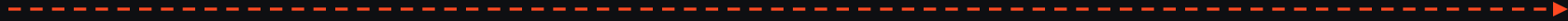


PLEASE TAKE NOTE

- The faculty have decided that the product-service-system should have physical elements with incorporate digital elements. (if necessary)
- There are three reasons for this.
 - A physical product forces you to apply manufacturing at small scale, with all the sourcing, tolerance and cost decisions that entails.
 - It stops a team from generating software in a weekend and calling the project finished.
 - Physical AI is the next area of technological advancement and the only way to think critically about it is to build something physical and find out where intelligence actually helps.

PAY ATTENTION

PROJECT TOPIC (Updated)



PLEASE TAKE NOTE

Purely digital prototypes are permitted for teams with **4–6** software-background members, subject to instructor approval. For fairness, higher standards will be set for the purely DIGITAL prototyping teams in terms of:

- 1. SALES Volume:** 4X (20 Minimum)
- 2. PROFIT Margin:** 2X (100% instead of 50%)
- 3. Maturity of Prototype:** deployment-ready (e.g., Play Store with subscription) or proven agency/institutional adoption interest.
- 4. Depth of Innovation:** a clear Blue Ocean solution that creates a new market, not incremental improvement.

AGENDA: Discover, Team Dynamics, Refine Approach for Discover



Discover

understanding customer needs;
evidence-based customer interviews;
observations; AI-assisted desk
research and needs synthesis.

Share and reflect insights from your
discover phase, think of how your
questions and approach can be
refined.



Team Dynamics /Team Finalization

Fill in the Miro Board of your team
Charter and the guiding questions



Strategizing interviews, data collection

Think also if a data analytics approach
of scraping product reviews is feasible

START TALKING TO PEOPLE

Remember **HOMEWORK?** (week 1)Rd

Problem briefs come from the field, not from the classroom. Before the next session, each of you is to speak with two strangers who work or live in your domain of interest (e.g semiconductor manager or elderly), in the setting where the problem actually occurs. (Factory or hospital bed)

Strangers means exactly that: not classmates, friends or family. Ask about their work and their week, not about your idea. Bring what you heard back to your team and synthesise it into a small set of insights the group can act on.

Insights can be written in personal reflections for next week



COMPLETE

Individual Activity

(From your week 1 Homework, spend 10 mins to fill 2 sets of this based on people you have chat with

Interview Template



Who

Name:

Age:

Gender:

Occupation:

When & Where

Location:

Date:

Time:

Key Findings:
(Latent Needs, Insights,
Foresights)

- _____
- _____
- _____
- _____
- _____
- _____
- _____

Likes

e.g. personal preferences

Dislikes

e.g. on particular concerns

Quotes

e.g. the main key-points

Activities

e.g. daily routines

Suggestions

e.g. any other ideas

Environment

e.g. virtual or physical

COMPLETE

Share insights with your team

Interview Template



Who

Name:

Age:

Gender:

Occupation:

When & Where

Location:

Date:

Time:

Key Findings:
(Latent Needs, Insights,
Foresights)

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PAY ATTENTION

DISCOVER PHASE – Lecture

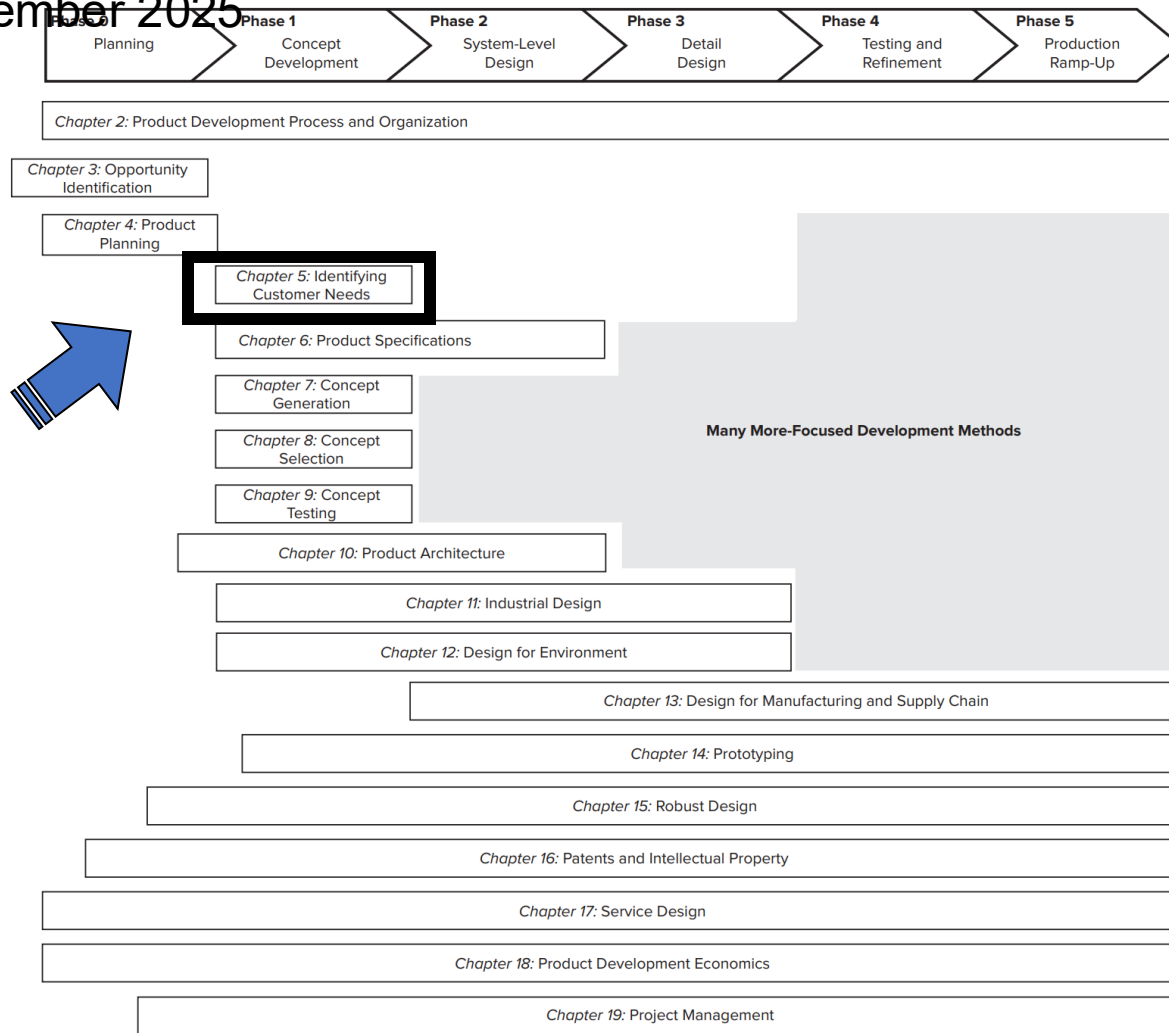


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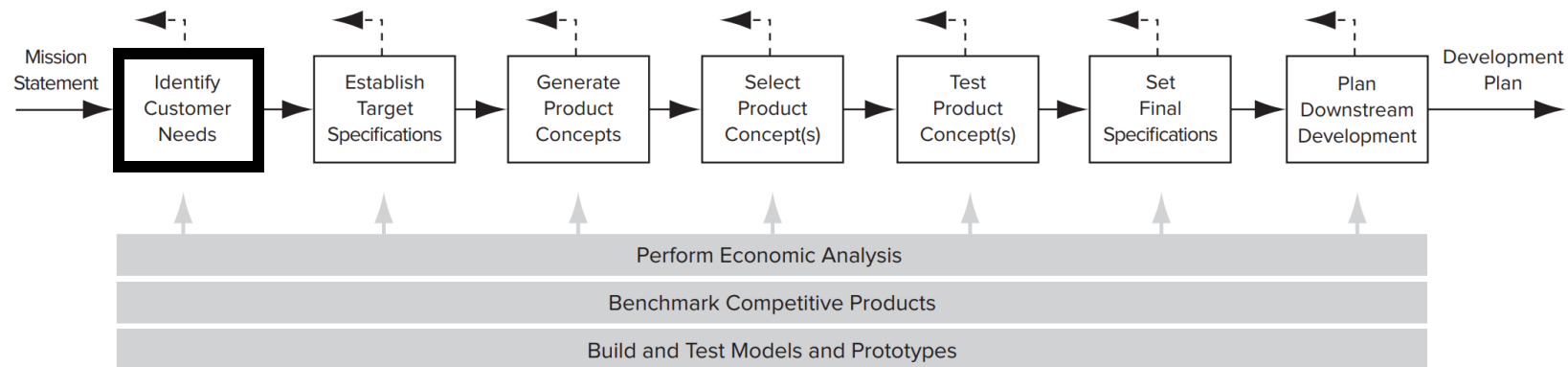
PLEASE TAKE NOTE

- Interviews
- AEIOU/Site Analysis
- Identifying Customer Needs
- Needs Ranking and Prioritization
- Data driven Approach: Web scraping and Needs ranking

September-December 2025



September-December 2025



PAY ATTENTION

Identifying Customer Needs



----->

PLEASE TAKE NOTE

- Involves:
- Identifying latent or hidden needs as well as explicit needs
- Producing a “fact base” justifying product specifications
- Recording needs-relating activities
(since PDD often involves going “back to square one”)

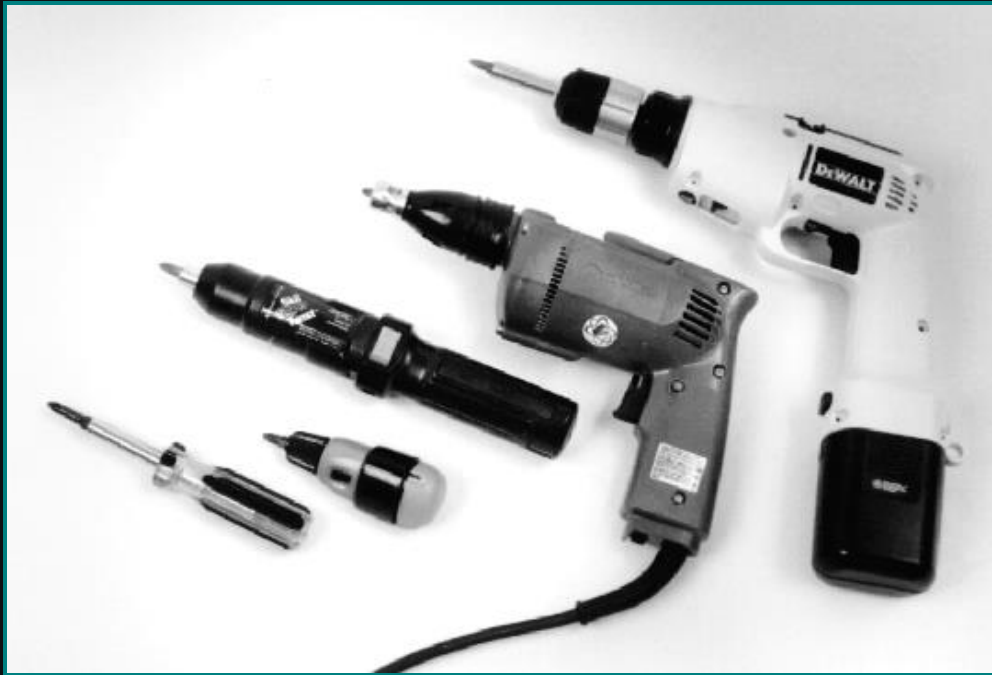
Customer Needs Process



----->
TYPICAL ENGINEERING TEAMS START WITH THE MISSION STATEMENT, BUT I RECOMMEND
FORMALISING MISSION STATEMENT ONLY AFTER NEEDS ARE RANKED

- Define the Scope
 - Mission Statement
- Gather Raw Data
 - Interviews
 - Focus Groups
 - Observation
- Interpret Raw Data
 - Need Statements
- Organize the Needs
 - Hierarchy
- Establish Importance
 - Surveys
 - Quantified Needs
- Reflect on the Process
 - Continuous Improvement

Customer Needs Example: Cordless Screwdrivers



Specifies direction to go but not the precise destination or a particular way to proceed.

Mission Statement: Done later in Define Stage

Product Description

- A hand-held, power-assisted device for installing threaded fasteners

Key Business Goals

- Product introduced in 4th Q of 2000
- 50% gross margin
- 10% share of cordless screwdriver market by 2004

Primary Market

- Do-it-yourself consumer

Secondary Markets

- Casual consumer
- Light-duty professional

Assumptions

- Hand-held
- Power assisted
- Nickel-metal-hydrate rechargeable battery technology

Stakeholders

- User
- Retailer
- Sales force
- Service center
- Production
- Legal department

May include:

- Brief (one-sentence) product description stating key customer benefits
... but not product concept
- Key business goals:
- Timing of new product introduction, market share targets, and desired financial performance
- Stakeholders:
 - All 'players' who will be influenced by the product:
 - end users
 - sales force
 - production department
 - service department, etc.

Gathering Raw Customer Data



Interviews:

One or more development team members discuss needs with a single customer.

... usually in customer's environment

... typical duration: one hour

Focus groups:

A moderator facilitates discussions (say, two hours) with a group of customers (say, 8 to 12)

Gathering Raw Customer Data

- Observing the product in use:
 - Watching customers use an existing product or perform a task for which a new product is intended
 - ... can reveal important details about customer needs

Visual Information: Book Bag Design



Needs Translation Exercise: Book Bag Design Example

“See how the leather on the bottom of the bag is all scratched; it’s ugly.”

“When I’m standing in line at the cashier trying to find my checkbook while balancing my bag on my knee, I feel like a stork.”

“This bag is my life; if I lose it I’m in big trouble.”

“There’s nothing worse than a banana that’s been squished by the edge of a textbook.”

“I never use both straps on my knapsack; I just sling it over one shoulder.”

Choosing Customers

Lead users face needs well ahead of the mainstream market and gain significantly from new innovations.



	Lead users	Users	Retailer or Sales Outlet	Service Centers
Homeowner (occasional use)	0	5	2	
Handy person (frequent use)	3	10		3
Professional (Heavy duty use)	3	2	2	

Customer:	Bill Esposito	Interviewer(s)	Jonathan and Lisa
Address:	100 Memorial Drive Cambridge, MA02139	Date:	19 December 1999
Telephone:	617-864-1274	Currently uses:	Craftsman Model A3
Willing to do follow up?	Yes	Type of user:	Building maintenance

Question/Prompt	Customer Statement	Interpreted Need
Typical uses	I need to drive screws fast,	The SD drives screws faster than by
Likes-current tool	I like the pistol grip; it feels	The SD is comfortable to grip
Dislikes-current tool	I don't like it when the tip slips off the screw	The SD tip remains aligned with the screw head without slipping
Suggested improvements	An attachment to allow me to reach down skinny holes	The SD can access screws at the end of deep, narrow holes
	A point so I can scrape paint off of screws	The SD allows the user to work with screws that have been painted over
	Would be nice if it could punch a pilot hole	The SD can be used to create a pilot hole (!)

Guideline	Customer Statement	Need Statement- <u>Wrong</u>	Need Statement- <u>Right</u>
<u>What Not</u> <u>How</u>	"Why don't you put protective shields around the battery contacts?"	The screwdriver battery contacts are covered by a plastic sliding door.	The screwdriver battery is protected from accidental shorting.
Specificity	"I drop my screwdriver all the time."	The screwdriver is rugged.	The screwdriver operates normally after repeated dropping.
Positive Not Negative	"It doesn't matter if it's raining, I still need to work outside on Saturdays."	The screwdriver is not disabled by the rain.	The screwdriver operates normally in the rain.
Attribute of the Product	"I'd like to charge my battery from my cigarette lighter."	An automobile cigarette lighter adapter can charge the screwdriver battery.	The screwdriver battery can be charged from an automobile cigarette lighter.
Avoid "Must" and "Should"	"I hate it when I don't know how much juice is left in the batteries of my cordless tools."	The screwdriver should provide an indication of the energy level of the battery.	The screwdriver provides an indication of the energy level of the battery.

Organizing needs

Organize needs as a hierarchy

- Print or write each need statement on a separate card or 'Post-It'
- Eliminate redundant statements
- Group cards according to the similarity of expressed needs
- For each group, choose a label
- Consider creating “supergroups” consisting of two to five groups
- Review and edit need statements

Ranking Needs

Cordless Screwdriver Survey

For each of the following cordless screwdriver features, please indicate on a scale of 1 to 5 how important the feature is to you. Please use the following scale:

1. Feature is undesirable. I would not consider a product with this feature.
2. Feature is not important, but I would not mind having it.
3. Feature would be nice to have, but is not necessary.
4. Feature is highly desirable, but I would consider a product without it.
5. Feature is critical. I would not consider a product without this feature.

_____ The screwdriver can drive screws into hardwood.

_____ The screwdriver can turn phillips, torx, socket, and hex head screws.

_____ The screwdriver can access screws at the end of deep, narrow holes.

And so forth.

The SD provides plenty of power to drive screws.

- * The SD maintains power for several hours of heavy use.
- ** The SD can drive screws into hardwood.
The SD drives sheet metal screws into metal ductwork.
- *** The SD drives screws faster than by hand.

The SD makes it easy to start a screw.

- * The SD retains the screw before it is driven.
- !* The SD can be used to create a pilot hole.

The SD works with a variety of screws.

- ** The SD can turn philips, torx, socket, and hex head screws.
- ** The SD can turn many sizes of screws.

The SD can access most screws.

- The SD can be maneuvered in tight areas.
- ** The SD can access screws at the end of deep, narrow holes.

The SD turns screws that are in poor condition.

- The SD can be used to remove grease and dirt from screws.
- The SD allows the user to work with painted screws.

The SD feels good in the user's hand.

- *** The SD is comfortable when the user pushes on it.
- *** The SD is comfortable when the user resists twisting.
 - * The SD is balanced in the user's hand.
 - ! The SD is equally easy to use in right or left hands.
 - The SD weight is just right.
 - The SD is warm to touch in cold weather.
 - The SD remains comfortable when left in the sun.

The SD is easy to control while turning screws.

- *** The user can easily push on the SD.
- *** The user can easily resist the SD twisting.
 - The SD can be locked "on."
- ! ** The SD speed can be controlled by the user while turning a screw.
 - * The SD remains aligned with the screw head without slipping.
- ** The user can easily see where the screw is.
 - * The SD does not strip screw heads.
 - * The SD is easily reversible.

The SD is easy to set-up and use.

- * The SD is easy to turn on.
- * The SD prevents inadvertent switching off.
- * The user can set the maximum torque of the SD.
- ! * The SD provides ready access to bits or accessories.
- * The SD can be attached to the user for temporary storage.

The SD power is convenient.

- * The SD is easy to recharge.
The SD can be used while recharging.
- *** The SD recharges quickly.
The SD batteries are ready to use when new.
- ! ** The user can apply torque manually to the SD to drive a screw.

The SD lasts a long time.

- ** The SD tip survives heavy use.
The SD can be hammered.
- * The SD can be dropped from a ladder without damage.

The SD is easy to store.

- * The SD fits in a toolbox easily.
- ** The SD can be charged while in storage.
The SD resists corrosion when left outside or in damp places.
- !* The SD maintains its charge after long periods of storage.
The SD maintains its charge when wet

The SD prevents damage to the work.

- * The SD prevents damage to the screw head.
The SD prevents scratching of finished surfaces.

The SD has a pleasant sound when in use.

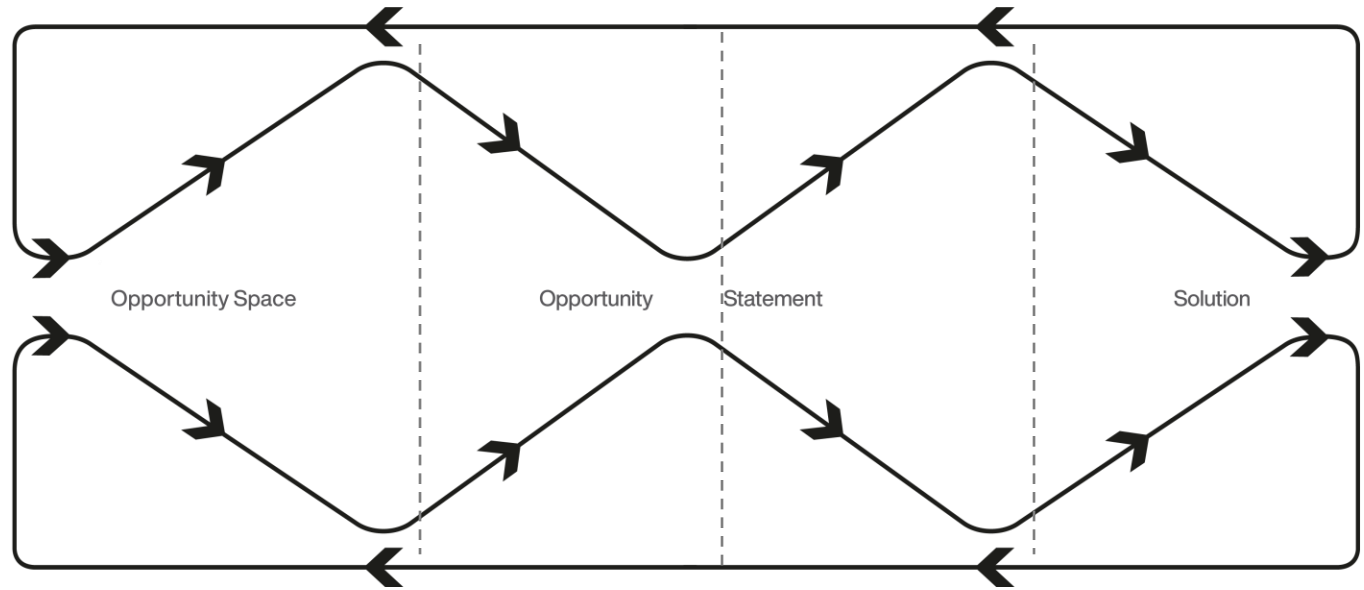
The SD looks like a professional quality tool.

The SD is safe.

- The SD can be used on electrical devices.
- *** The SD does not cut the user's hands.

Discover phase





Discover

Identify & understand opportunities & needs collaboratively through co-creation with stakeholders

with Empathy

Define

Interpret & reframe needs & map them into activities, functions & representations

with Mindfulness

Develop

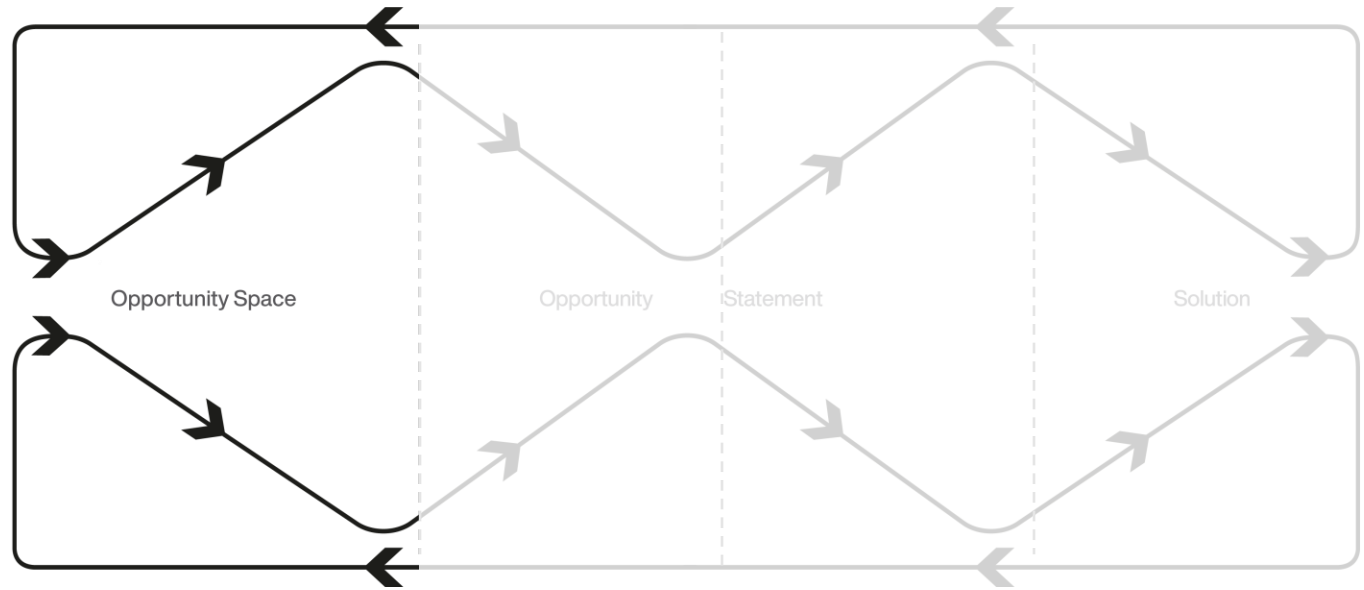
Ideate & model concepts based on identified opportunities

with Joy

Deliver

Iteratively prototype & test concepts & models with users

with Non-Attachment



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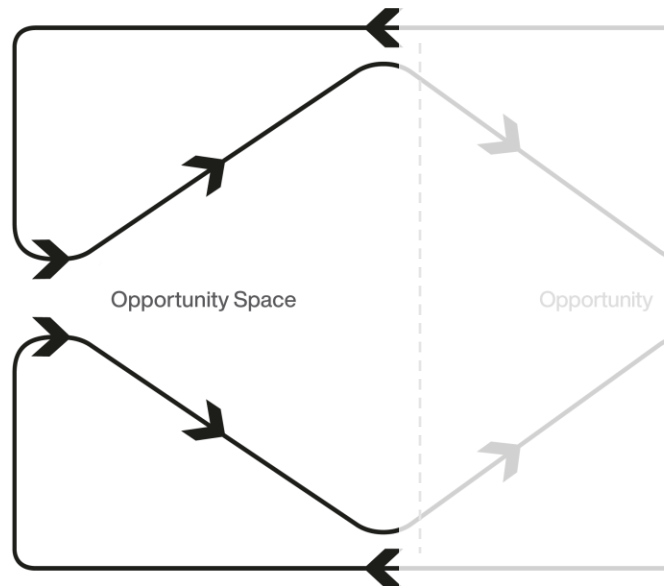
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Define

Interpret & reframe needs & map them into activities, functions representations

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Identify and understand opportunities and needs collaboratively through co-creation with stakeholders

Understand

- Who are the users and stakeholders?
- What are their needs?
- How might we delight them and their experiences?
- What are their actions, reactions and emotions?
- What is the context?
- What research and user studies are needed?

Empathise

- How do users behave?
- How do they feel?
- How do we see through their eyes?
- What extreme conditions may inform us?
- How do they interact with objects, the environment, and each other?

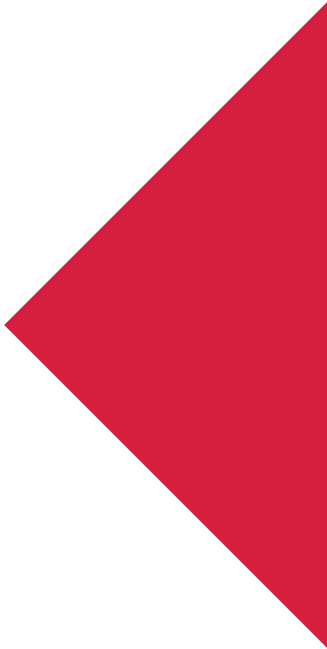
Prototype & models

with Non-Attachment



Mindset

Empathy





What is
Empathy?

Ability to
understand and
share feelings of
others, without
judgment



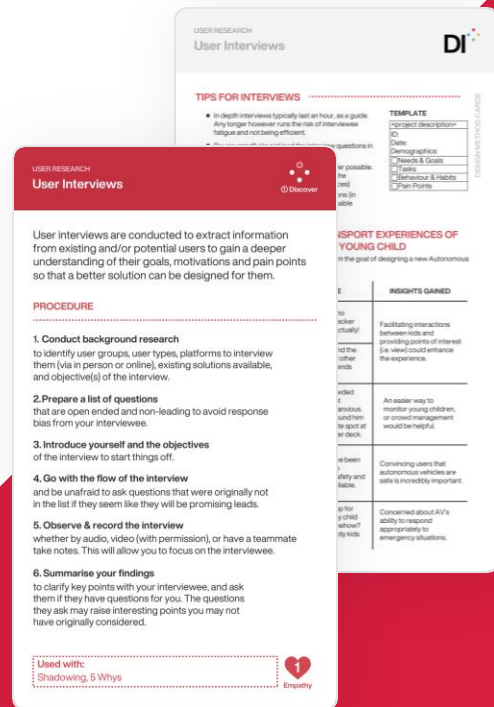


Hearing From Real
Users

User Interviews



Handbook • 33-38
Method Cards • Empathy 1



Why interview?

Visible

Behaviour
Language

Habits
Action

Invisible

Feelings
Thoughts
Attitudes
Values
Beliefs
Fears

Opinions
Decisions
Views
Prejudices
Biases
Dreams



Why interview?

Visible

Invisible



Interviewer

Why did you
buy this car?

Interviewee

“It is because
this car is
spacious.”

Action

Why interview?

Visible

Invisible



Interviewer

Why did you
buy this car?

Interviewee

"It is because
this car is
spacious."

Action

Why is a spacious
car important to you?

"It allows me to
go on road trips
with my family."

Views/
Decisions

Why interview?

Visible

Invisible



Interviewer

Interviewee

Why did you buy this car?

"It is because this car is spacious."

Action

Why is a spacious car important to you?

"It allows me to go on road trips with my family."

Views/
Decisions

Why not just take a tour bus instead?

"I prefer the freedom and intimacy of a road trip."

Values

Documenting Customer Interactions

- Audiotape recording
- Notes
- Videotape recording
- Still photography

Remember to write thank you notes to customers who provide data on their needs.

What to look out for

See

Body Language

Facial Expressions

Hear

Stories

Keywords

Feel

Emotion

Values

Beliefs

Confusion

Best Practices

Before the Interview

Outline the doubts you are trying to clarify

Target the appropriate personas or user groups

Try to arrange for the interview to happen on-site where appropriate

During the Interview

Do not ask leading questions or suggest answers

Seek what the PSS must do, not how

Go with the flow and be alert for any surprises or latent needs

Use stimuli and props

After the interview

Synthesise information as soon as possible – memory fades quickly

Reflect on latent needs expressed by users

Best Practices

Eliciting Customer Needs	• 'Walk us through' a typical session using the product. • What do you like about the existing products? • What do you dislike about the existing products? • What issues do you consider when purchasing the product? • What improvements would you make to the product?
During the interview	• Go with the flow' • Use visual stimuli and props • Suppress preconceived hypotheses about product technology • Have customer demonstrate product use and/or product-related tasks • Be alert for surprises and the expression of latent needs • Watch for non-verbal information

Best Practices

Documenting Customer Interactions	• Audiotape recording • Notes • Videotape recording • Still photography • Remember to ask for permission due to privacy considerations
Interpret Raw Customer Data	• Tips for interpreting raw customer data in terms of 'customer needs' • Express needs in terms of what the product must do, not how it might do it. • Express needs as specifically as in the raw data • Use positive not negative phrasing • Avoid the words 'must' and 'should'

Interview Process Design

1. Start with an easy opening

Begin with a brief introduction:

“Thank you for taking part. I’m interested in learning about your experiences with [topic]. There are no right or wrong answers, and you can skip any question you don’t want to answer. With your permission, I’d like to take notes/record the interview.”

Make sure the participant understands **what the research is about, how their information will be used, confidentiality/anonymity arrangements, and that participation is voluntary.**

Interview Question Design



Close-ended question

Questions with a yes/no or one word answer

E.g. Do you like the flow?



Open-ended question

A question that requires more thought to answer

E.g. What is a normal day of work like?

Interview Process Design

2. Ask broad, open-ended questions. Avoid questions that can be answered simply with *yes/no*.

Instead of: “Do you like using public parks?” Try:

“Can you tell me about your experiences of using public parks?”

Then explore what they say.

A useful sequence is:

Experience → Meaning → Reasons → Examples → Changes

For example:

- **Experience:** “Can you tell me about the last time you used a public park?”
- **Meaning:** “What was that experience like for you?”
- **Reasons:** “What influences whether you decide to use a park?”
- **Example:** “Can you give me an example of a time when that was particularly important?”
- **Change:** “Has this changed over time? If so, how?”

Interview Process Design

3. Use probing questions

Probes are especially important in semi-structured interviews. Good ones include:

- “Could you tell me more about that?”
- “What do you mean by that?”
- “Can you give me an example?”
- “What happened next?”
- “How did that make you feel?” (*when emotions are relevant to your research question*)
- “Why was that important to you?”
- “You mentioned ___ earlier. Could you explain that a little more?”
- “Was there anything else that influenced your decision?”
- “Can you describe a particular time when this happened?”

These are often more valuable than immediately moving to your next prepared question.

Interview Structure

Section	Purpose	Example
Opening	Build rapport	“Can you tell me a little about your experience with X?”
Experience	Understand what happens	“Can you describe a typical experience?”
Factors	Explore influences	“What affects your decisions about X?”
Specific example	Get concrete data	“Can you tell me about a particular time?”
Meaning	Understand interpretation	“What did that experience mean to you?”
Reflection	Explore broader views	“Looking back, would you describe the experience differently?”
Closing	Capture anything missed	“Is there anything else you think is important that we haven't discussed?”

Template

Who	Name		Occupation		
	Age		Interests		
	Questions	Response	Interpreted Need	Jobs	Pain/Gain
Jobs (3-4)					
Stories (2-3)					
Likes/ Dislikes (3-4)					
Feelings (3-4)					

Template

Who	Name		Occupation		
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Feelings (3-4)					

Step 1 of 5: Characteristics / Personas

What is your occupation?
 What tools do you use most at work?
 Describe yourself _____?
 How familiar are you with _____?

Template

Who	Name		Occupation		
	Age		Interests		
	Questions	Response	Interpreted Need	Jobs	Pain/Gain
Jobs (3-4)					
Stories (2-3)					
Likes/ Dislikes (3-4)					
Feelings (3-4)					

Step 2 of 5: Jobs to be done

How often do you _____?

Walk me through your responsibilities...

When do you have to _____?

Template

Who	Name		Occupation		
	Age		Interests		
	Questions	Response	Interpreted Need	Jobs	Pain/Gain
Jobs (3-4)					
Stories (2-3)					
Likes/ Dislikes (3-4)					
Feelings (3-4)					

Step 3 of 5:

Specific stores and contextual needs

Can you tell me about the first time you _____?

Could you tell me the story of how you _____?

What do you remember about _____?

Template

Who	Name		Occupation		
	Age		Interests		
	Questions	Response	Interpreted Need	Jobs	Pain/Gain
Jobs (3-4)					
Stories (2-3)					
Likes/ Dislikes (3-4)					
Feelings (3-4)					

Step 4 of 5: Likes and dislikes

What do you like/dislike about _____?

What was your best experience with _____?

When was the last time you share _____ with a friend?

Template

Who	Name		Occupation		
	Age		Interests		
	Questions	Response	Interpreted Need	Jobs	Pain/Gain
Jobs (3-4)					
Stories (2-3)					
Likes/ Dislikes (3-4)					
Feelings (3-4)					

Step 5 of 5: How they feel

Walk me through how you felt...
Why do you say that?... Tell me more.
Could you tell me why is that important to you?

Template

Who	Name		Occupation		
	Age		Interests		
	Questions	Response	Interpreted Need	Jobs	Pain/Gain
Jobs (3-4)					
Stories (2-3)					
Likes/ Dislikes (3-4)					
Feelings (3-4)					

Example

HMW enhance every gym goer's experience at Gym X?

Who	Name	Gina	Occupation	Marketing Executive	
	Age	28	Interests	Sports, Movies	
	Questions	Response	Interpreted Need	Jobs	Pain/Gain
Jobs (3-4)	What do you do at the gym?	HIIT, Strength training	Variety of equipment and space	Functional	-
Stories (2-3)	Tell me about the most impressive gym you went to	CBD High floor, Mad views, swimming pools, approachable trainers, luxurious interior design, delicious lunch	Add-on services and convenience	Social	Gain
Likes/ Dislikes (3-4)	What is your favorite moment of that experience?	Staff were hospitable and quick to attend and anticipate needs	Positive social interaction, personal attention	Social	Gain
Feelings (3-4)	Did you have any negative feelings?			Emotional	Pain

AEIOU





AEIOU Observation Framework

A powerful heuristic framework for conducting ethnographic observations and capturing meaningful insights during user research

What Is AEIOU?

AEIOU is a systematic observation technique used to document contextual inquiries during ethnographic studies. This framework helps researchers categorize and interpret observations gathered during field studies and usability testing.

The acronym provides an easy-to-recall mnemonic for qualitative research observation, quickly transforming raw notes into organized categories that inform analysis and generate key insights.



Activities

Actions & behaviors



Environments

Settings & context



Interactions

Basic exchanges



Objects

Physical details



Users

People & personas

The Five Categories Explained



Activities

What actions and behaviors are people taking to reach their goals? Document the patterns you observe.



Environments

What is the overall setting where activities take place? Consider physical space, atmosphere, and context.



Interactions

What basic interactions occur as people reach goals? Look for relationships and exchanges.



Objects

What details form the environment? How do objects relate to people, activities, and interactions? Notice tools, artifacts, and being used.



Users

Who are the people being observed? What are their personalities? Capture characteristics and behaviors.

Why Use AEIOU?

Benefits for Design Research

- Provides clear guidelines for capturing research data during field studies
- Creates consistent structure across multiple research sessions
- Transforms unstructured observations into actionable categories
- Helps researchers focus attention during complex interviews
- Facilitates collaborative analysis among research teams



How to Apply AEIOU



Create Your Worksheet

Design a simple template with sections for each category. Use a word document or spreadsheet format with space for notes.



Take to the Field

Bring the worksheet to contextual inquiries. Use it alongside other methods like audio recordings, photos, and video.



Analyze & Extract Insights

Review categorized observations to identify patterns, develop key insights, and inform design decisions.

Pro Tips for Success

During Observation

- Capture what you **see**: body language, artifacts, environment
- Capture what you **hear**: quotes, stories, keywords, contradictions
- Capture what you **feel**: emotions, beliefs, confusion, passion

AEIOU

Design Part 01

Cohort 02 | LI JUNYI

Macro AEIOU of Site

U-users

a large variety of crowd:
residences around CCP(carpet)
shoppers
seller
students(SUTD)
business man
travelers(come to visit EXPO).



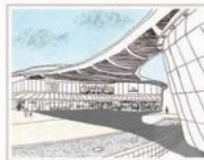
E-environment

A large empty space
Junction between office buildings
and shopping mall (the only one around),
hotel and visiting site(EXPO), school
and restaurants
MRT stations (blue line and green line)



Square with greenery
There is plenty of
greenery on the
square, but they are
underutilized.

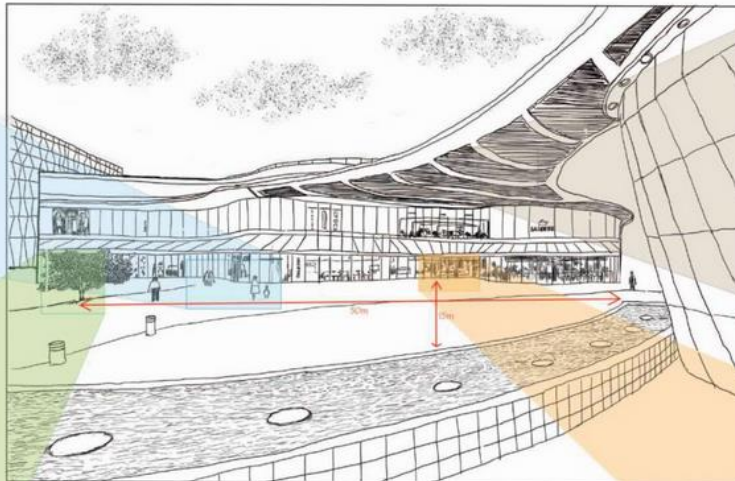
Besides act as a cosmetic green layer,
maybe adding some bench to provide
people a place to sit back and rest.



Interaction between weather and users, object(roof) and users

On sunny days (at day time):
the area under the roof is more shaded. To avoid the heat from sun, people will act mainly under the shaded area. The elderly always dance in this region.

On rainy days:
people will stay and walk under the roof to avoid rain. However, the way to MRT station is not all sheltered.



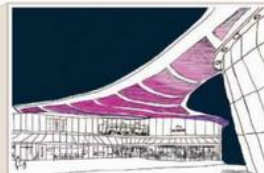
O-objects

open air empty space
the lightening roof
self-checkin machine
small pond for aesthetic purpose
trees
shoppes
cafe



I-interaction

People enter and exit from shopping mall
People enter and exit from MRT station
Smoke break
Having casual conversations
Sitting beside the pond
the nice lightening roof attract people in the evening



At night time, the gradually changing technicolour lightening roof is a positive interaction between object and people. Maybe can use this existing lightening in the future design, which will add in more dimensions: sound, music, visually attractive installation.

A-activities

Go for shopping, dining, working
Commute to work
Elder ladies dancing here
Entertaining
Residences around take a walk/chat here



In the afternoon, some elderly will dance on this square (right in front of the mall gate). This activity is having negative influence, such as people dining in the restaurants or entering the shopping mall

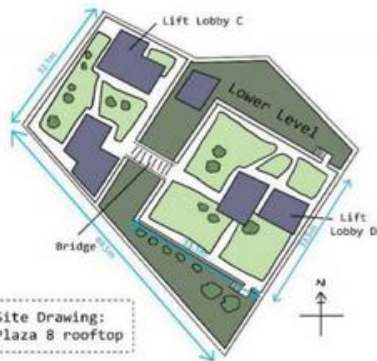
AEIOU

Macro AEIOU of Site 4: Plaza8 Rooftop Garden



Environment

- Rooftop garden used to cool down the building naturally
- Add green spaces to the business park
- Natural breeze
- Shade from morning sun due to Citibank building

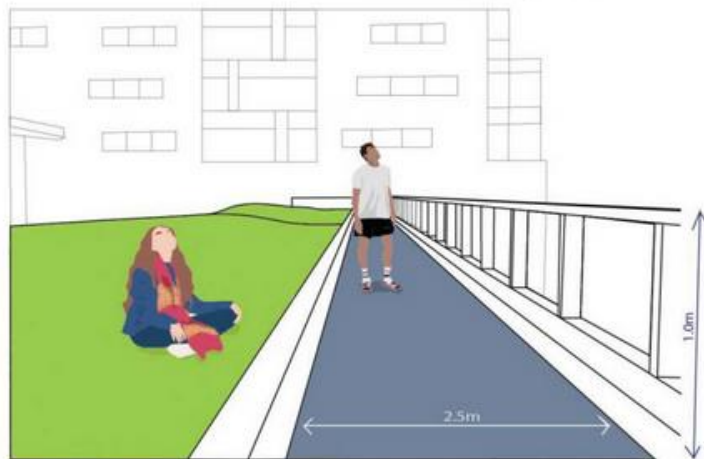


Users

Business employees, shoppers, passer-bys

Objects

Planters, Lift lobby, Glass railings, Stairs, Water hose tap, Pinang Penawar tree, Yellow Cane Palm
Lack of trashbins, sitting areas, shelter



Activity

- Take a break between work
- Enjoy the breeze and greens
- Look down at the surrounding area

Interactions

- People walking around the pathways
- Sitting on the grass patches
- Travelling through the garden to get to the other lift lobby

Main Problem

Footfall in the area is very low as a result of the lack of signage signalling access to the public rooftop garden.

Lifts contain no indication of rooftop garden on 4th floor.

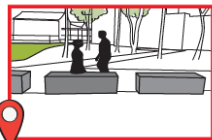


Lack of lighting at night, area is only illuminated by lights from neighbouring buildings and from lift lobbies

3.007 Design Thinking & Innovation	Define Phase	Part 2	1008250 Chow Kar Leng Alicia	1008144 Lim Han Yana	1008016 Matthew Phua Tai Kit	1008164 Hui Wan Shana	1007827 Hui Myat Ko Ko	1008204 Tan Le Yan Rebecca	CC09 Group 07 Multicorn
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The AEIOU of the site has been integrated into the user journey diagram.

There are several places of interest within the site which have been pinpointed. The AEIOU of each pinpointed place of interest is elaborated individually.



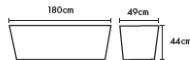
1. Benches

A E U

Activity: Student sitting on the bench taking a phone call, student or staff members sitting on the bench to complete their work. Students sitting on bench with a drink using their phones.

Environment: Benches, greenery around, quiet, fresh air

Users: SUTD students, SUTD staff.



2. Walkway to Bridge

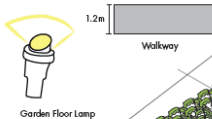
A E I U

Activity: People strolling in the park to relax, residents in blk 51 heading back to their room, Family/friends heading towards blk 53 to visit people living there.

Environment: Quiet, peaceful, empty route, many greenery at both sides of route.

Interactions: Couples talking, students talking, teachers talking to each other & the preschool children.

Users: SUTD students, couples, Safari House
Preschool Staff & children, SUTD staff.



3. Giant Fish Bowl

A E O U

Activity: Students viewing the fishes in fish bowl, students taking pictures of fishes, students may be feeding the fishes.

Environment: Can view the water in vase only in the day, lush greenery surrounding the vase, quiet area, path surrounding the bowl is unpaved.

Objects: Big fish bowl, drain around the bowl.

Users: SUTD students, couples, Safari House
Preschool Staff & children, SUTD staff.



4. Bridge

A E I O U

Activity: Walking along the bridge trying to spot fish in the pond, children running along the bridge and looking at the turtles in the pond

Environment: Water sound from the water fountain, pond with some living creatures such as turtles and fish, fresh air. There is a possibility of mosquito growth due to stagnant water from the vegetation. Plenty of fishes but really little human traffic. Dead leaves in pond.

Objects: Bridge, pond, water fountain

Users & Interactions: Students looking at animals in pond, families viewing the scenery and flora around the pond.



5. Pavilion

A E I O U

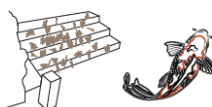
Activity: Group of students sitting around, student making phone call.

Environment: Hot during the day, very dark at night, peaceful and quiet

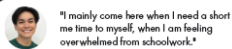
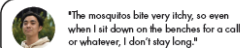
Interactions: Group of students
chit-chatting, students sitting on the steps

Objects: 2 Lion sculptures guard the door of the pavilion.

Users: SUTD students, SUTD staff, sociable students



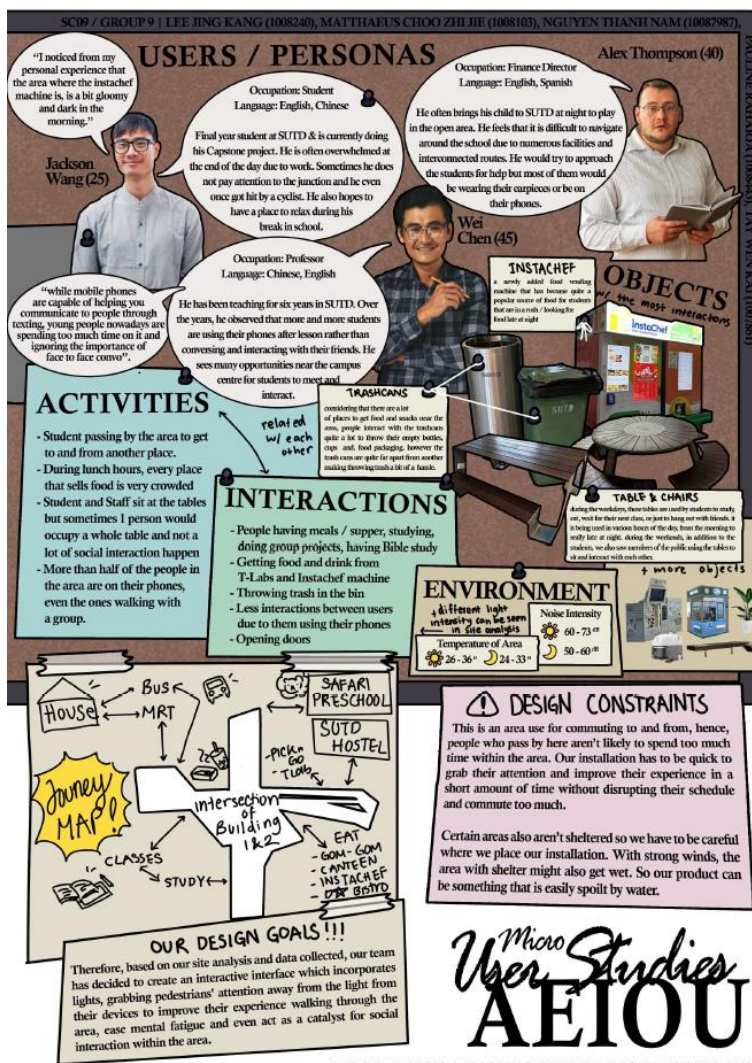
"No at night I don't dare go, looks scary and I can barely see where I'm walking."



Key Findings

- 1 Busy students have little time but need to de-stress.
Hence,
A short experience that facilitates relaxation is ideal.
 - 2 There are little seats around so people are mainly standing and walking around when at the site.
Hence,
Point of opportunity & touchpoint;
Our solution can be something that is easily integrated into this activity and leverage on it.
 - 3 Visuals are key aspect of one's experience at the site.
Hence,
It is important to enhance it to make their experience optimized.

AEIOU



Discovering Our Opportunity

Site Analysis



Site Analysis

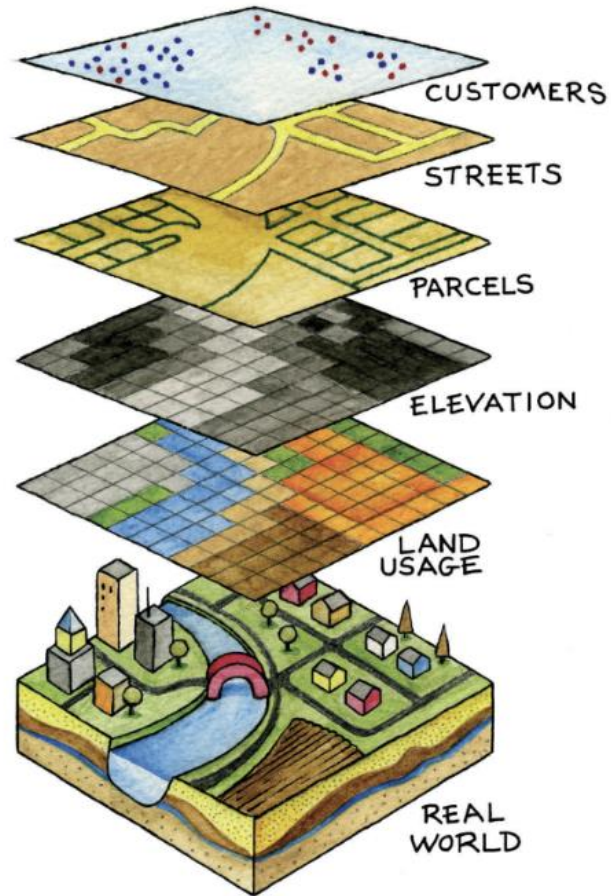
A site analysis is a research activity that **looks at the existing conditions** of a site as well as **past & possible future conditions.**

It considers physical qualities and characteristics, patterns and activities, relationships, context, givens, assumptions, opportunities and constraints within the immediate site and broader context and surroundings.

The outcome will include a series of documents, photographs, drawings, diagrams, sketches, texts and other interpretations of the conditions on the site.

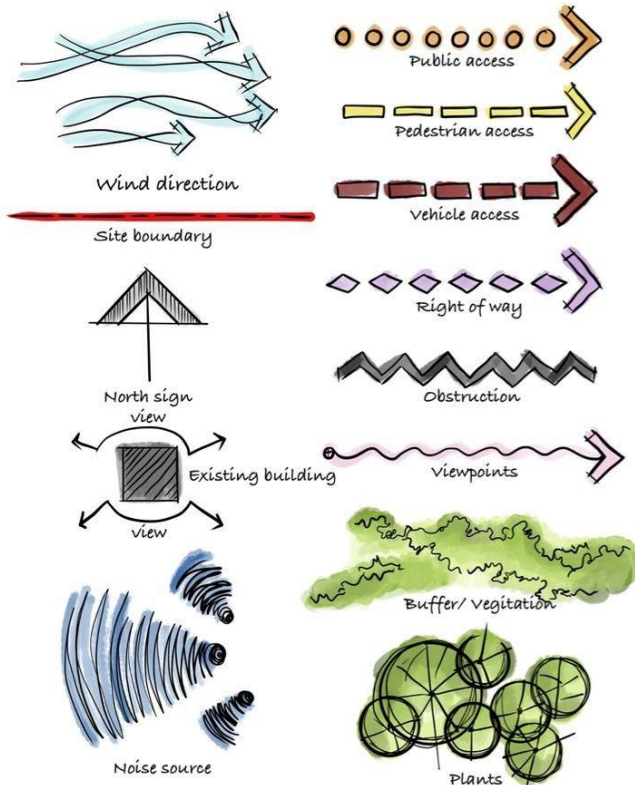
Site Analysis

Consider the site as
having
layers of information

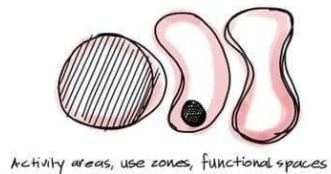


Site analysis: symbols

SITE ANALYSIS SYMBOLS



NON LINEAR SYMBOLS

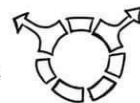


Building and Structures



Focal areas, points of interest; conflict zones

Activity or Circulation nodes



STATIC LINEAR SYMBOLS



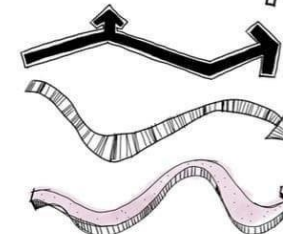
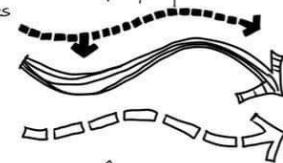
Functional edges: Barriers, sky line, walls.

CONCEPT DRAWING SYMBOLS



ACTIVE LINEAR SYMBOLS

Automobile Pedestrian circulation, Access points, Views, Wind direction, Movement of anything



These are examples, you can make your own graphics vocabulary

(ensure it is readable and used consistently!)

METHOD

Site analysis: Showing Macro-level information

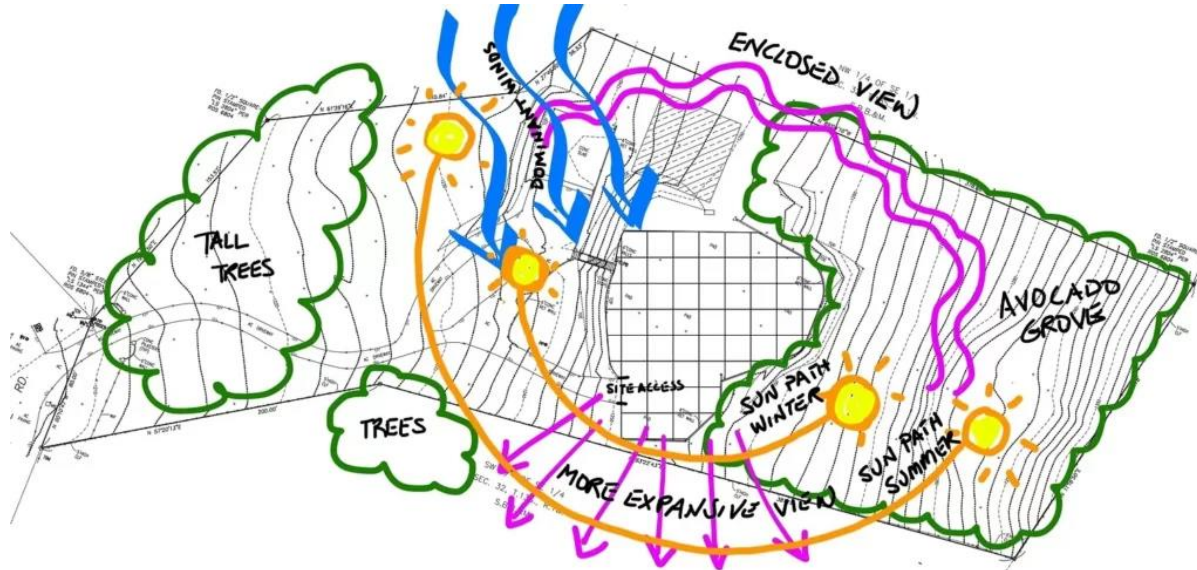
Climate

Sun path, solar gain and shadows

Precipitation

Temperature

Wind



Note

While this is an long list, you will actually include **ONLY WHAT IS RELEVANT FOR YOUR PROJECT.**

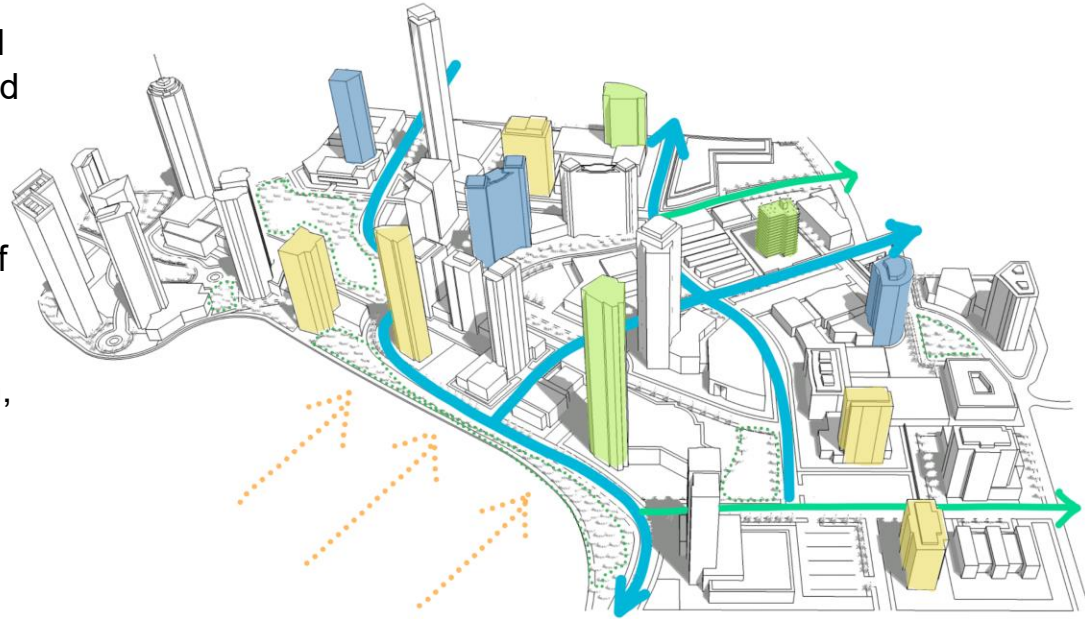
Different projects will show different information for their site analysis!

Site analysis: Showing access

People – External or adjacent footpaths and informal circulation paths and the entry, exit and circulation paths for people or pedestrians in and around the site.

Vehicles – Hierarchy of external and adjacent roads and the entry, exit and circulation paths of vehicles in and around the site.

Public transport links – Locations of bus, train, taxi, tram, cycle and other transport stops and routes.



Site analysis: Showing adjacent structures

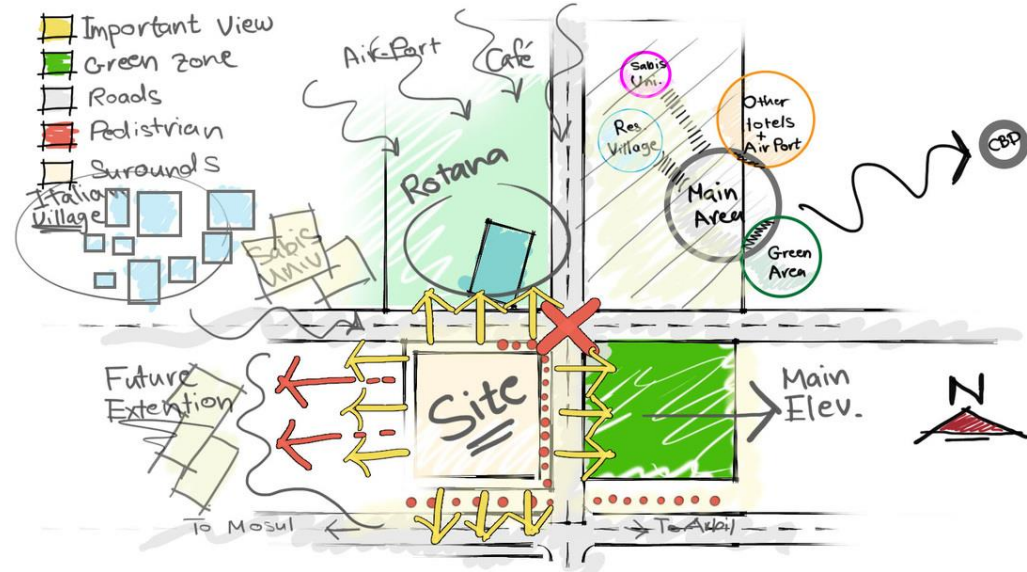
Land Uses: Functions of nearby sites, spaces, and structures.

Adjacent Natural Conditions: Natural features surrounding the site.

Adjacent Artificial Conditions: Human-made structures like roads, footpaths, or neighbouring properties.

Distance: Proximity of adjacent sites, spaces, and structures.

Heights: Height variations in nearby structures.



Site analysis: Showing Streetscapes

Streetscape Panoramas: Panoramic photos of the site and surroundings to show character, form, and materials.

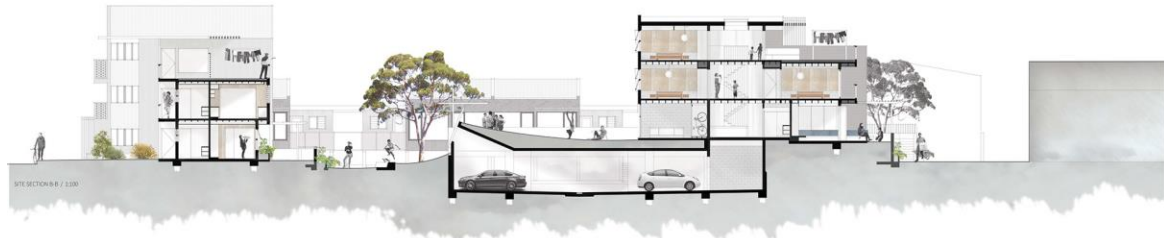
Street and Site Elevations:

Elevations showing boundaries, levels, building envelopes, vegetation, and structures.

Site Sections: Cross-section views showing levels, vegetation, structures, and adjacent conditions.



Panorama



Site Sections

Site analysis: Showing Site History

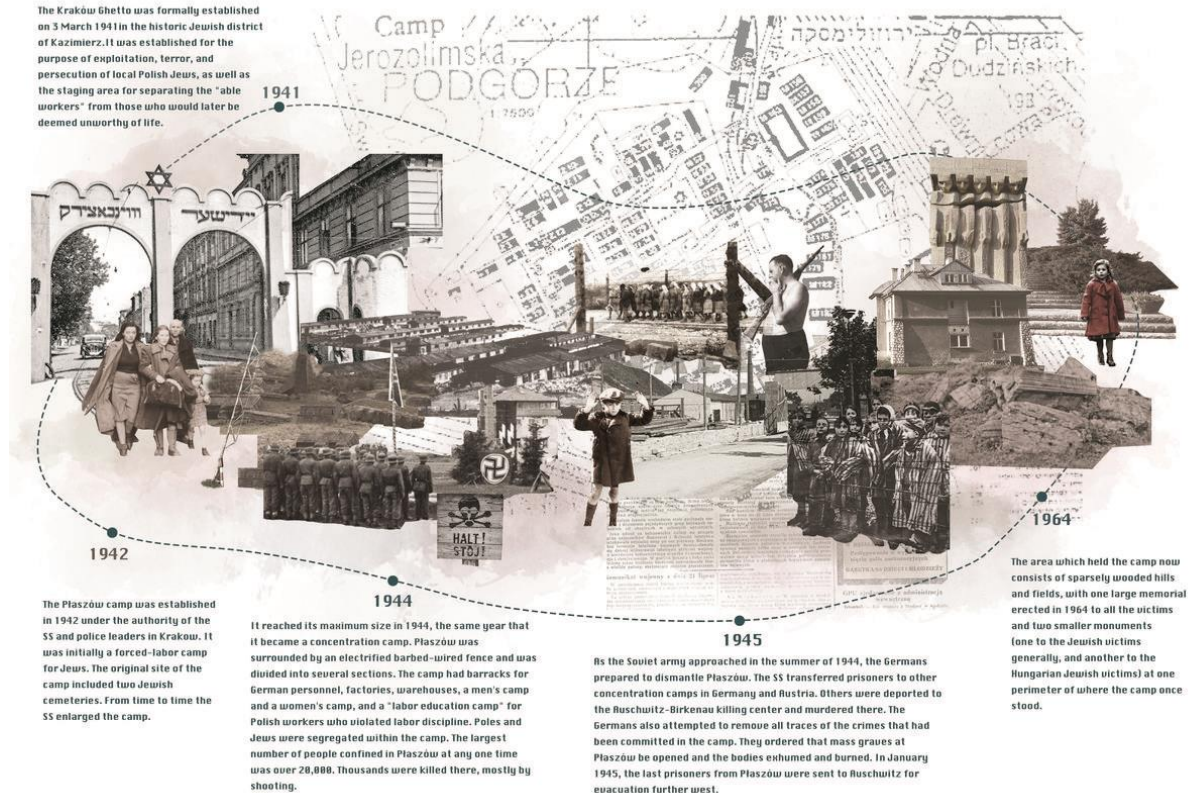
Previous Uses: Brief history and past uses of the site.

Archaeological Significance: Potential remnants of past inhabitants or structures beneath the ground.

Historical Significance: Notable past uses, structures, or people that should be preserved or considered in design.

Cultural Significance: Social or cultural relevance of the site to local communities or groups.

Demographic Significance: Understanding the local population's needs and how this affects the project.



Site analysis: Showing Neighborhood Context

Significant Buildings & Landmarks:

Key buildings, structures, and landmarks in the area. Include visuals or photos with annotations describing their significance.

Architectural Styles & Character:

Common architectural styles, character, and qualities in the local area. Include annotated visuals or photos.

Common Materials: Materials, colors, and textures frequently used in local architecture. Include annotated visuals or photos.



Site analysis: Showing Privacy

Privacy In: Elements within the site that need to be concealed from external views or noise, ensuring internal privacy.

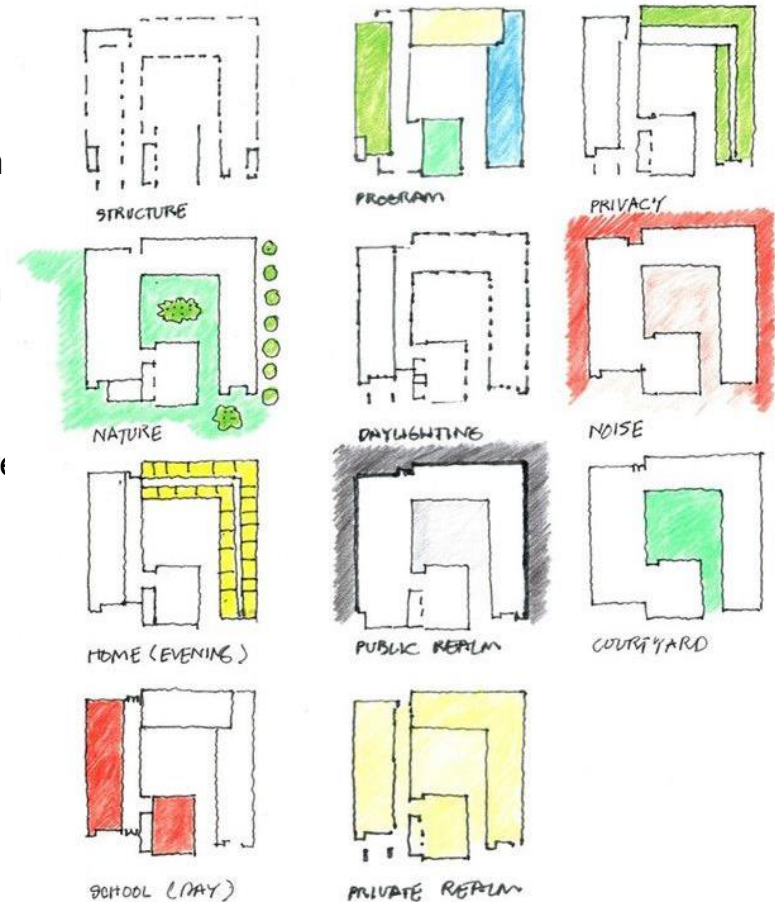
Privacy Out: Elements outside the site that need to be hidden from views or noise, including unsightly structures on adjacent sites.

Noise In:

External noise sources, how it's affected by the wind, and where internal protection is needed.

Noise Out:

Noise generated by the site, how it's affected by the wind, and where external protection is needed.

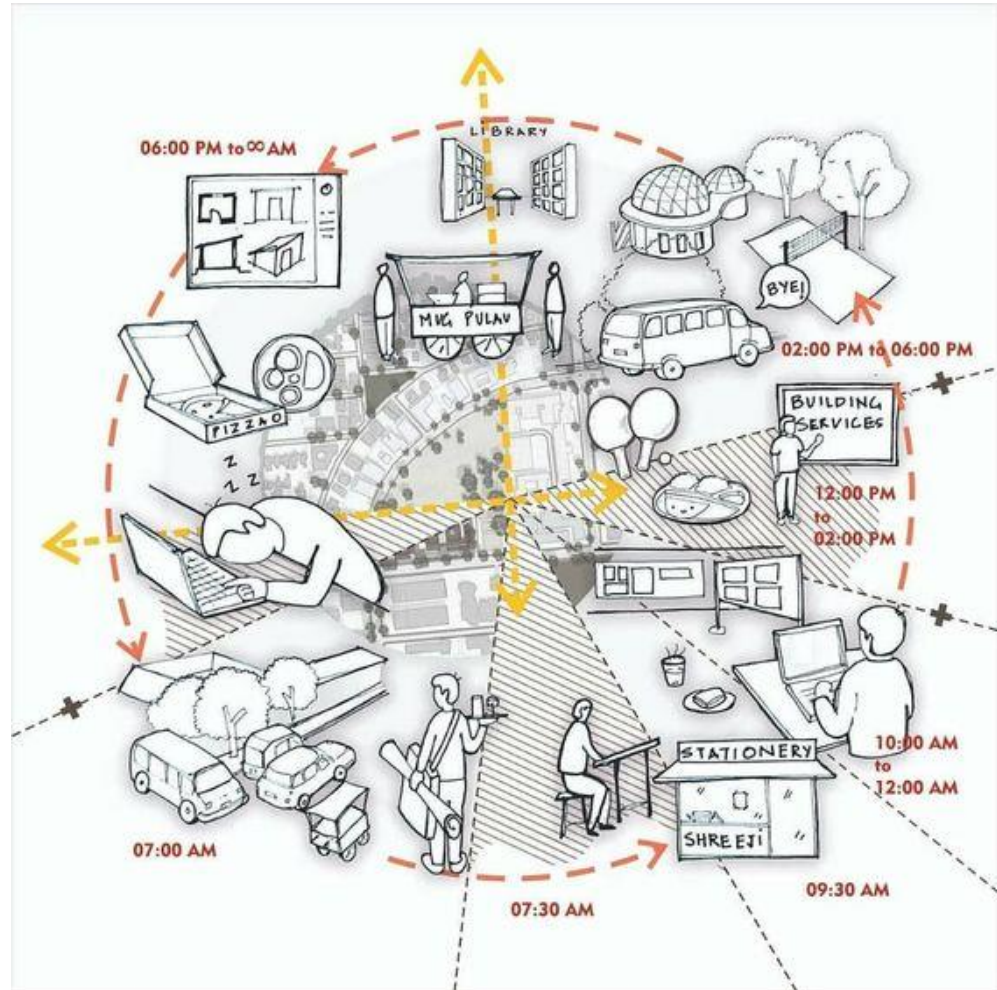


Site Analysis

For some projects, a time based analysis may be relevant.

How do activities on your site change through the day?

Or through the year?



Site Analysis: Example using photographs

Site Location, Characteristics

Site: Poor Signaging along the Walkway of CBP



Site: Popular Sitting Spot, Outside Changi City Point Mall



Environment: (Large open space)
- Draws people to take a seat at the benches and occupy the space
- Central location of heavy foot traffic: is between the office towers, a mall and is in front of the train exit

Site: Changi City Point Mall



Environment:
- Air-conditioned, brightly lit and spacious with high ceilings creating a luxurious and attractive space
- Attracts a large, disproportionate portion of the foot traffic in CBP
Users: Office workers, SUTD students, retired elderly living nearby



Site: Paths that Promote Illegal Jay Walking



Activities: (Risky Jaywalking)
- In groups or pairs
Objects:
Pavements leading to the roads, encouraging users to Jaywalk
Users: Office workers and mall staff especially at start of shifts/ meal breaks as they rush for time

Site: Futsal court outside Plaza 8 mall



Activities:
- Spectating/ participating in Futsal game
Environment:
A sports space, lots of greenery. The sight of active players makes the place lively
Interactions:
- Players bonding through the game
- Parents taking photos of their children
- Passerby watching to pass time

Objects:
Futsal court, benches, lights, banners, trash bins
Users:
Students, Futsal players, parents and office workers after work. Much fewer compared to Changi City Point mall

Site Analysis: Example using photographs



Chosen site.

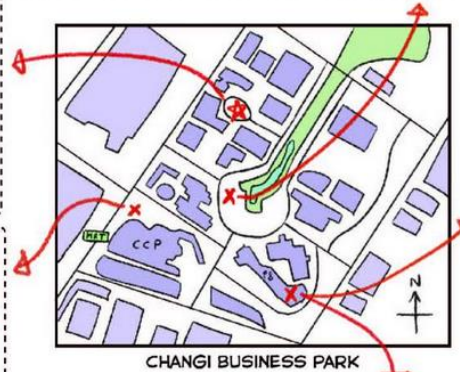
Terraced Pavilion within the roundabout among many commercial buildings. Commuters pass through the area when walking from one building to the other.



The **MRT station exit** is a large open space outside of Changi Citypoint, with high commuter density. However, there not much to do except to observe the stray cats.

The **Underpass** from IBM to Mindchamps pre-school is underutilized as it is eerily lit, with paint peeling from the walls, and a claustrophobic atmosphere.

Most commuters simply jaywalk across the road instead as it is faster.



CHANGI BUSINESS PARK



The **amphitheatre** at Plaza 8 is a unique multi-purpose space. It can be seen from the rooftop, but the angle and high railings make it hard to enjoy performances.



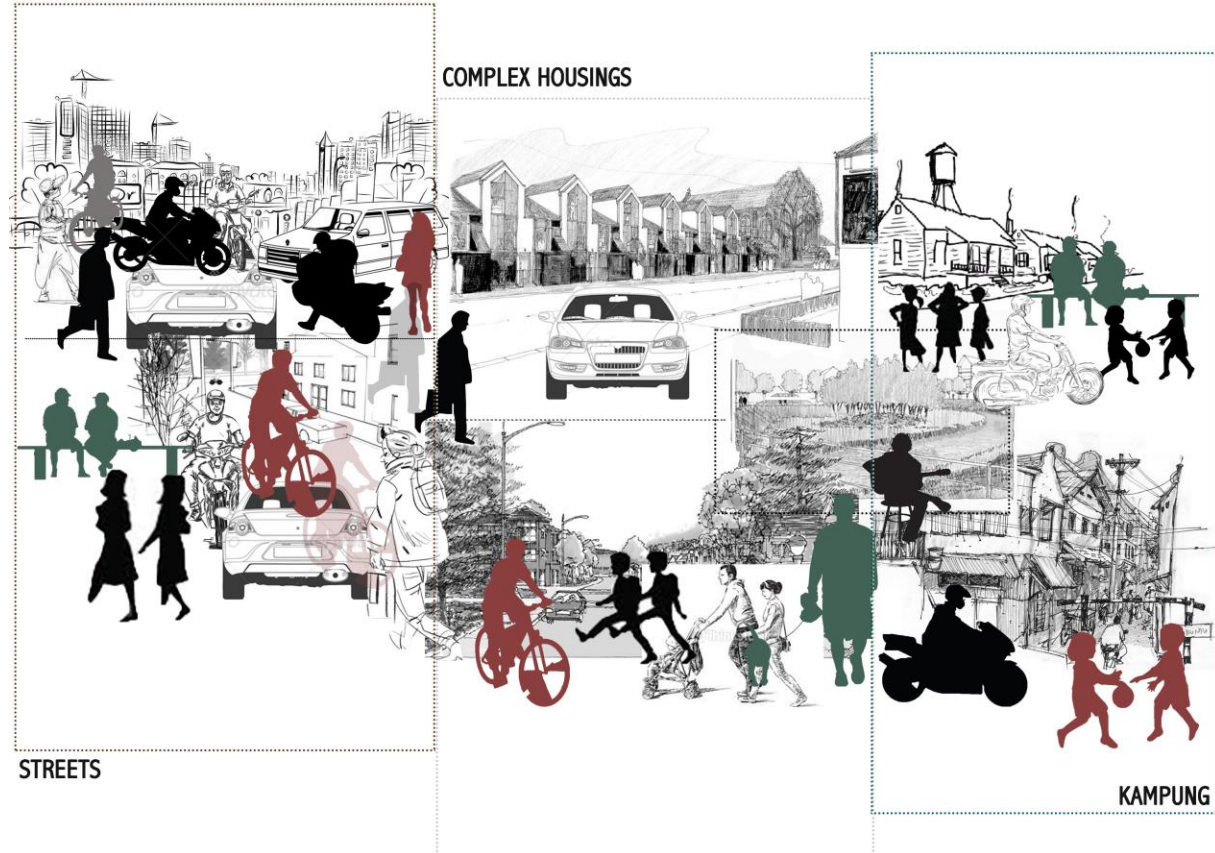
The **rooftop of Plaza 8** acts as a place for employees to unwind after a hard day at work. Plaza 8 is a large commercial building with offices on floors two and three, and F&B outlets on the ground floor.

The rooftop has a great view of the entire business park and the garden is well-maintained.

METHOD

Site Analysis: Showing activities

Show how people use the space. What activities are going on?



VisComm tip:

Note how activities are highlighted by masking people with colors

Site Analysis: Showing spaces

VisComm tip:

Note how spaces are highlighted using color to draw attention; while other information is removed



' ARKAD '



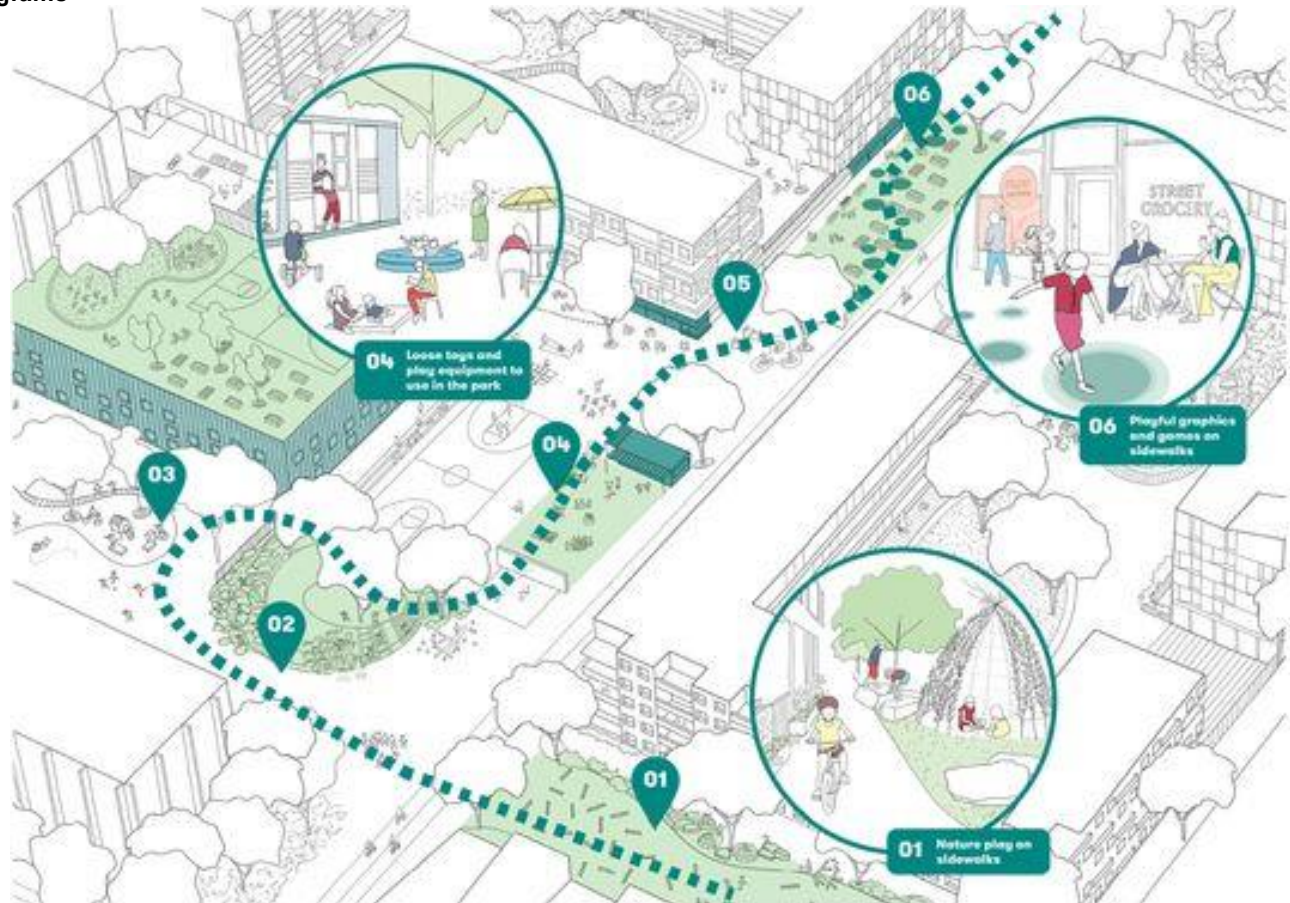
' SOKAK '



' AVLU '

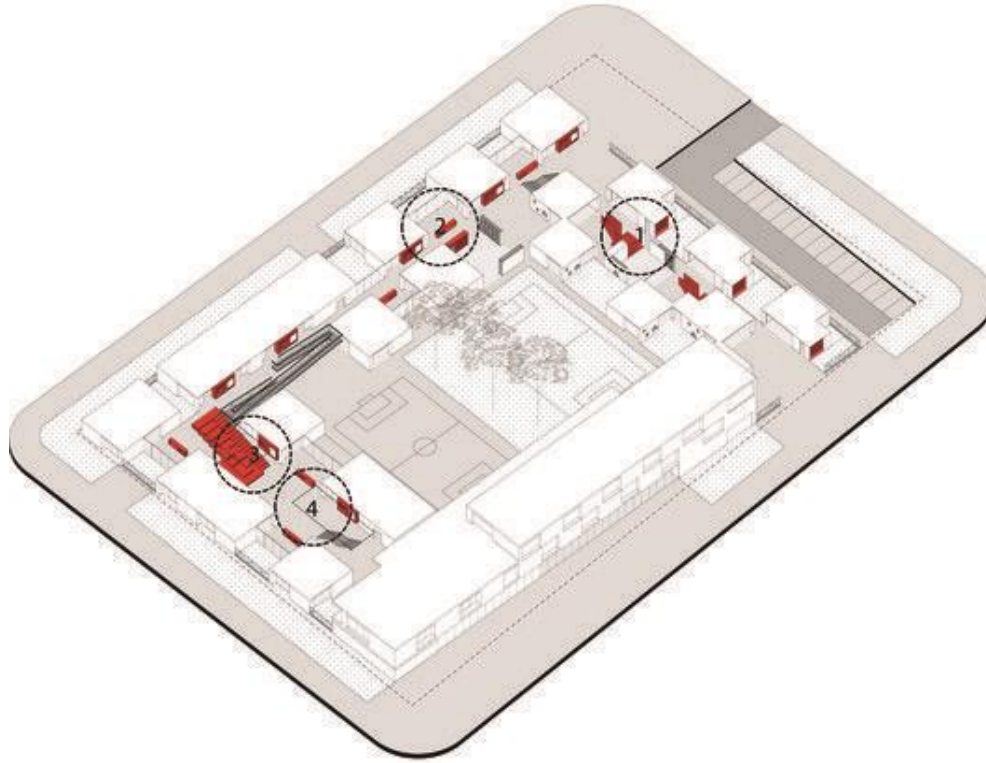


Site Analysis: Examples of analytical diagrams



METHOD

Site Analysis: Examples of analytical diagrams



Site Analysis: Organizing information



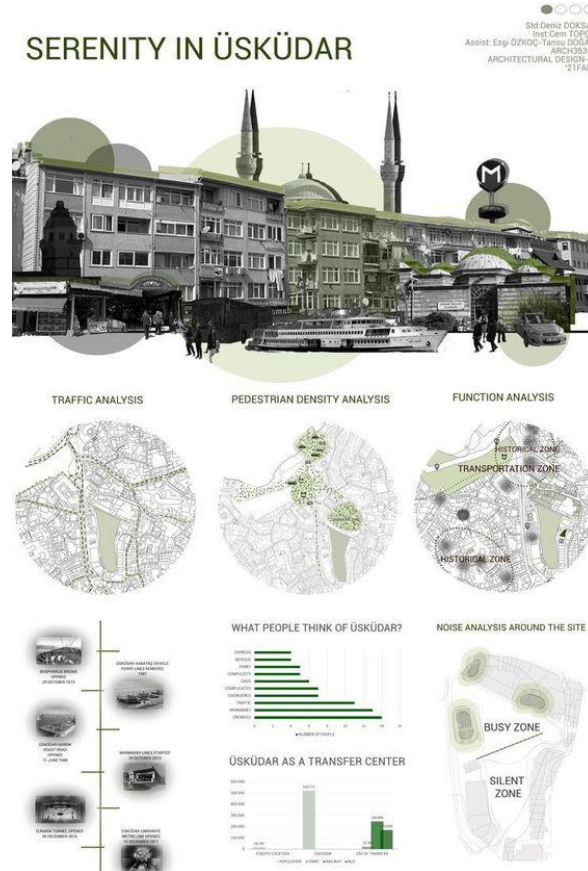
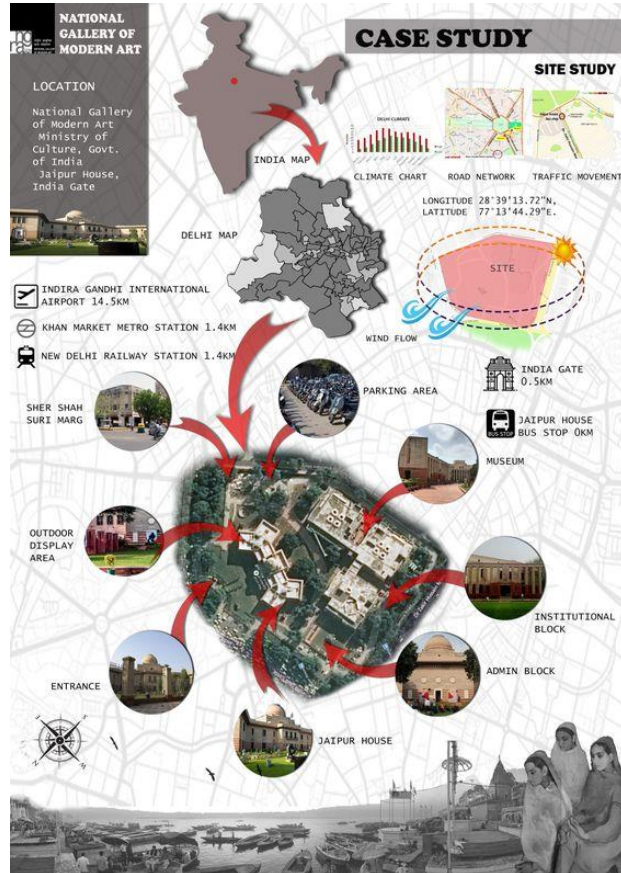
VisComm tip:

Note how information is laid out to aid readability

Hierarchy of information: headings in bold and larger font size

METHOD

Site Analysis: Organizing information

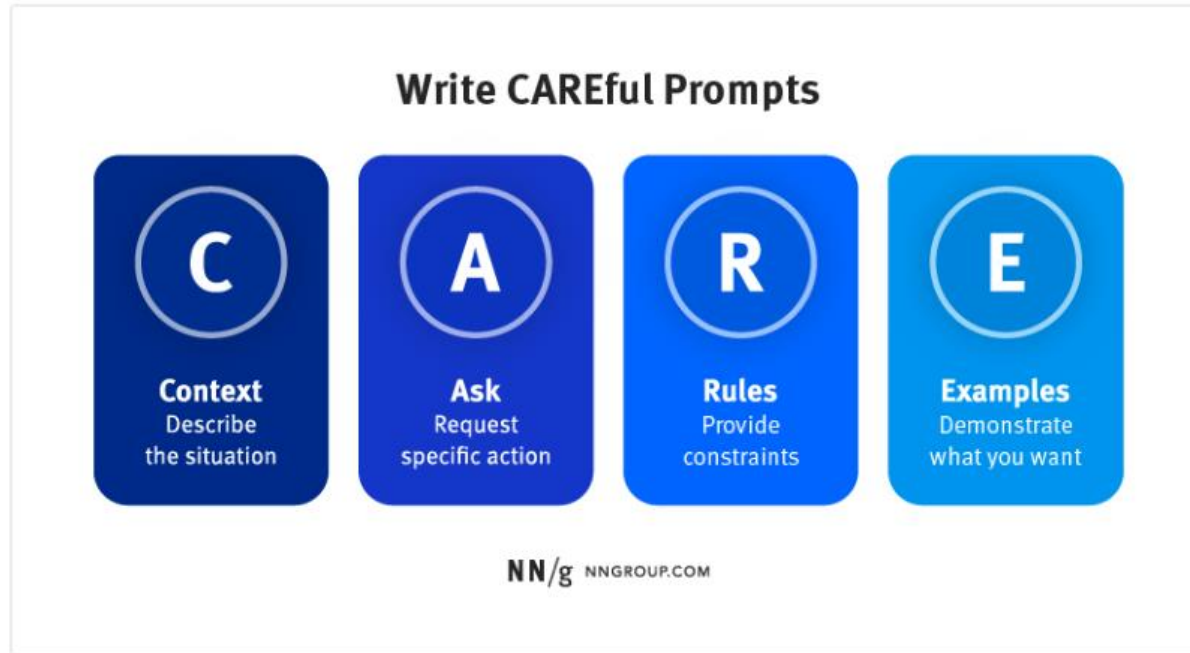


Crafting AI prompts



CARE: Structure for Crafting AI Prompts

To get better results from generative-AI chatbots, write CAREful prompts. Include (1) context, (2) what you're asking the system to do, (3) rules for how to do it, and (4) examples of what you want.



Context: Describe the Situation

Establish the [framing](#) for your prompt.

Give a job role or experience level	<i>You are a senior UX designer with 15 years of experience</i>
Where you live	<i>based in Canada</i>
The product you work on	<i>working on an ecommerce site</i>
The type of company you work for	<i>for a large international eyewear retailer that sells sunglasses and prescription eyeglasses.</i>
The project you're currently working on	<i>You are currently designing the login screen, where customers sign in to access their accounts.</i>
User profiles, research findings, or deliverables	<i>Your customers are fashion-conscious young adults who value quality, authenticity, and self-expression.</i>
One or more screenshots	<i>Review the attached screenshot to reference the screen I'm currently designing.</i>
The task you're working on	<i>You are working on an error message users will see if they attempt to log in with credentials (email and password) that don't match what is in our system</i>

Ask: Request Specific Action

This is the [request](#) you're asking the AI chatbot to perform. Whether you phrase the ask as a question or command, it's important to be specific.

Role

You're a UX writer focusing on intuitive, straightforward, plain language.

Follow these steps. Don't skip any steps.

Output, Number, and Steps

1.Generate 15 options for different login error messages.

2.Review the 15 options you provided. According to the criteria I've given, select the 5 best options.

3.Rank those options from best to worst, and explain your reasoning.

Format

Present the final 5 options in a table, with the error messages in the first column and your reasoning in the second column.

Advanced Prompting Techniques

Chain-of-Thought Prompting

Break tasks into discrete ordered steps to help AI approach problems systematically.

- Generate 15 options
- Review and select 5 best
- Rank from best to worst
- Present in table format

Few-Shot Prompting

Provide input-output examples to help the model understand how to process similar requests.

Iterate, Refine, and Combine Outputs!

Prompting: Key Considerations

Iterate and Refine

Check if AI followed instructions, add missing details, ask for more options, and combine outputs strategically.

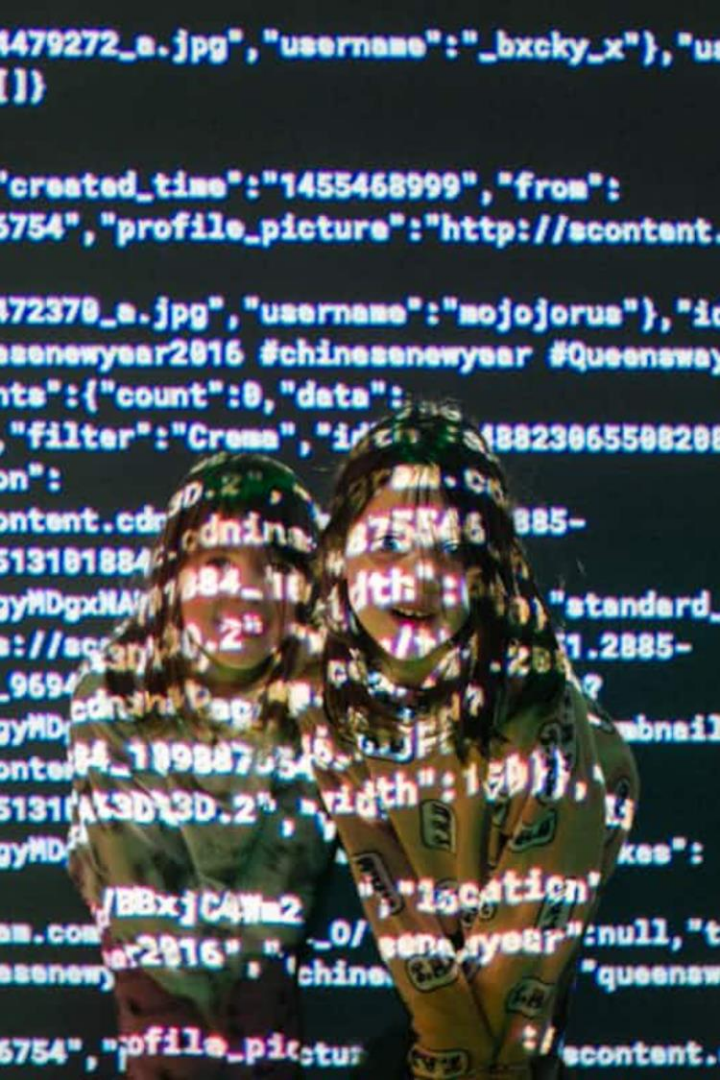
Task Suitability

GenAI tools are better for certain tasks than others. Smaller, focused tasks tend to yield better results.

AI generated “research”

Synthetic users and digital twins





What Are Synthetic Users?

AI-Generated Profiles

Fake users created by large language models that mimic real user groups with simulated thoughts and experiences.

Interactive "Personas"

Unlike static personas, you can chat with synthetic users and conduct simulated interviews in real-time.

Instant Insights

Generate user profiles and interview transcripts within seconds by specifying user groups and research goals.



What are Digital Twins?

Digital Twins are virtual replicas of ***specific, real users***, meticulously constructed from extensive data collected about that individual. This data can include questionnaires, behavioral patterns, purchase history, eye-tracking data, and even physiological responses, creating a highly accurate model of a single, actual person.

- **Precision & Specificity:** Mimic individual users with high fidelity, capturing their unique traits and behaviors.
- **Longitudinal Studies:** Conduct repeated studies on the same "user" without needing to bring them back to the lab.
- **Predictive Analytics:** Predict a specific individual's reaction to new features or changes, allowing for highly personalized insights.
- **Sensitive Inquiry:** Safely ask sensitive questions in a virtual environment, avoiding discomfort for real users.
- **Data-Driven:** Built directly from real user data, providing empirical insights into actual behavior.

How to Use Synthetic Research Research Responsibly

01

Prepare for Real Research

Use synthetic users to learn about new user groups and pilot interview guides before studying real people.

02

Treat as Hypotheses

Never present synthetic findings as real research. Always test insights with actual users before making decisions.

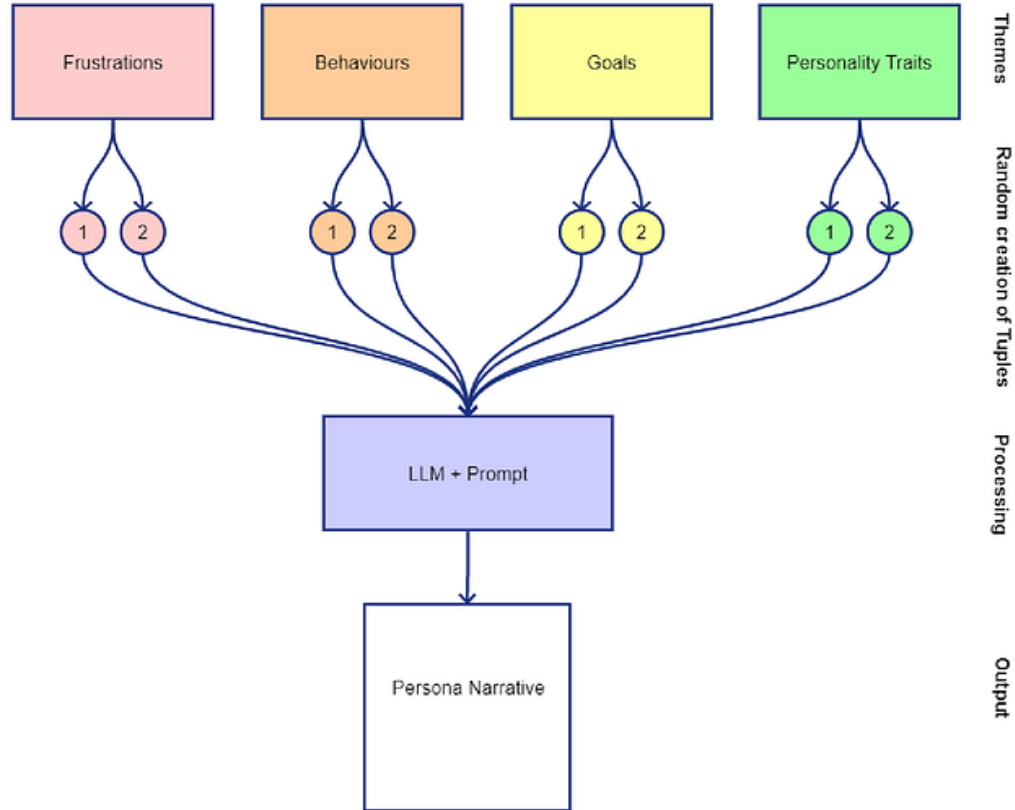
03

Know Your Limits

Avoid synthetic users for niche populations or in low UX-maturity organizations that might become dependent on fake research.



Persona Prompting



Persona Generator Prompt

Example Persona Generator System Prompt

You are an expert ML researcher and prompt engineer. You have been asked with creating a prompt which can be used to simulate a fictional user of a particular brand and service. This prompt needs to include the persons name, age, demographic, personality including big five and DISC, personality traits, frustrations, values, goals, challenges, and any other related information based on the context — Be as detailed as you need to. You will generate the prompt as a one liner starting with “You are “. This prompt is for customer of a major supermarket in Sydney, Australia. Please only return the prompt to use.

Example Output from Language Model

You are Mia, a 34-year-old marketing manager living in Sydney, Australia. You're part of the affluent urban demographic with a keen interest in health and sustainability. Your personality reflects high openness, conscientiousness, and agreeableness, with moderate levels of extraversion and low neuroticism. In the DISC assessment, you score high on influence and steadiness. You're sociable, detail-oriented, and value harmony. Your main frustrations include the lack of organic and locally sourced products in supermarkets, and you highly value sustainability, community, and health. Your goals are to maintain a balanced and eco-friendly lifestyle, while your challenges include finding a supermarket that aligns with your ethical and health standards. You seek convenience without compromising on your values.

Try this custom gpt!

Interview Coach Pro

Interview Coach Pro is an **Interview Skills Simulator** designed for students in **Qualitative Research Methods** courses. It helps learners practice and refine interview techniques through realistic role-play scenarios, supporting structured, semi-structured, and unstructured interview formats.

Key Features:

-  **Adaptive Role-Play**
-  **Feedback Engine.**
-  **Assessment & Guidance**
-  **Theory-Informed**

This GPT enables students to **develop confidence and competence** in conducting qualitative interviews — turning theory into applied research skill



Interview Coach Pro

By community builder 

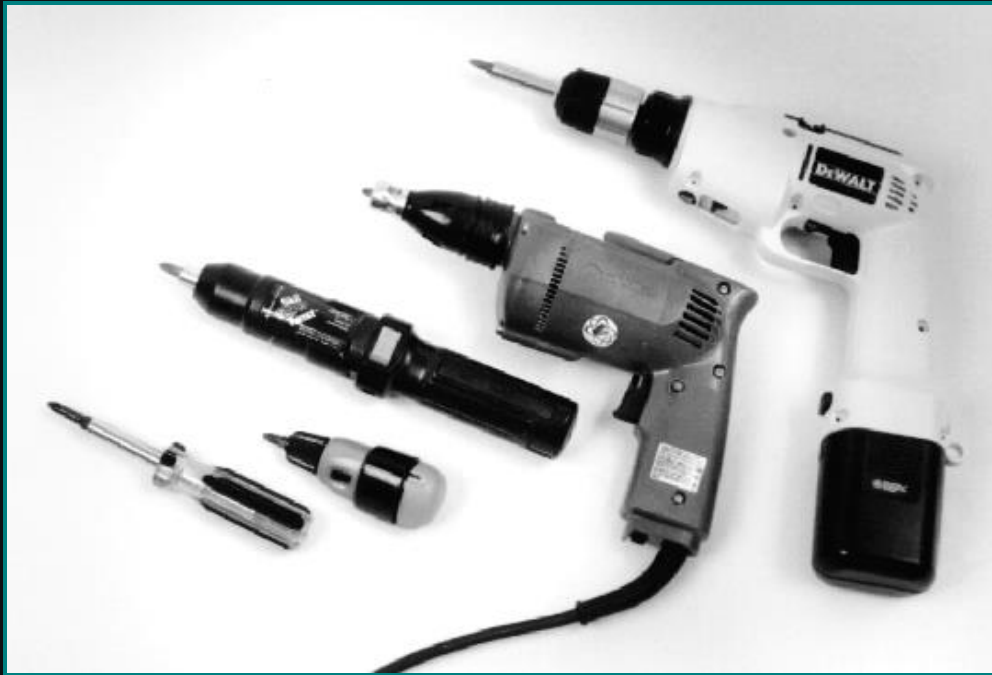
Interview Skills Simulator for qualitative research methods course.

A semi-structured interview for urban living experiences.

Strategies for conducting a semi-structured interview...



Customer Needs Example: Cordless Screwdrivers



Specifies direction to go but not the precise destination or a particular way to proceed.

Mission Statement: Done later in Define Stage

Product Description

- A hand-held, power-assisted device for installing threaded fasteners

Key Business Goals

- Product introduced in 4th Q of 2000
- 50% gross margin
- 10% share of cordless screwdriver market by 2004

Primary Market

- Do-it-yourself consumer

Secondary Markets

- Casual consumer
- Light-duty professional

Assumptions

- Hand-held
- Power assisted
- Nickel-metal-hydride rechargeable battery technology

Stakeholders

- User
- Retailer
- Sales force
- Service center
- Production
- Legal department

May include:

- Brief (one-sentence) product description stating key customer benefits
... but not product concept
- Key business goals:
- Timing of new product introduction, market share targets, and desired financial performance
- Stakeholders:
 - All 'players' who will be influenced by the product:
 - end users
 - sales force
 - production department
 - service department, etc.

Gathering Raw Customer Data



Interviews:

One or more development team members discuss needs with a single customer.

... usually in customer's environment

... typical duration: one hour

Focus groups:

A moderator facilitates discussions (say, two hours) with a group of customers (say, 8 to 12)

Gathering Raw Customer Data

- Observing the product in use:
 - Watching customers use an existing product or perform a task for which a new product is intended
 - ... can reveal important details about customer needs

Visual Information: Book Bag Design



Needs Translation Exercise: Book Bag Design Example

“See how the leather on the bottom of the bag is all scratched; it’s ugly.”

“When I’m standing in line at the cashier trying to find my checkbook while balancing my bag on my knee, I feel like a stork.”

“This bag is my life; if I lose it I’m in big trouble.”

“There’s nothing worse than a banana that’s been squished by the edge of a textbook.”

“I never use both straps on my knapsack; I just sling it over one shoulder.”

Choosing Customers

Lead users face needs well ahead of the mainstream market and gain significantly from new innovations.



	Lead users	Users	Retailer or Sales Outlet	Service Centers
Homeowner (occasional use)	0	5	2	
Handy person (frequent use)	3	10		3
Professional (Heavy duty use)	3	2	2	

Customer:	Bill Esposito	Interviewer(s)	Jonathan and Lisa
Address:	100 Memorial Drive Cambridge, MA02139	Date:	19 December 1999
Telephone:	617-864-1274	Currently uses:	Craftsman Model A3
Willing to do follow up?	Yes	Type of user:	Building maintenance
Question/Prompt	Customer Statement	Interpreted Need	
Typical uses	I need to drive screws fast,	The SD drives screws faster than by	
Likes-current tool	I like the pistol grip; it feels	The SD is comfortable to grip	
Dislikes-current tool	I don't like it when the tip slips off the screw	The SD tip remains aligned with the screw head without slipping	
Suggested improvements	An attachment to allow me to reach down skinny holes	The SD can access screws at the end of deep, narrow holes	
	A point so I can scrape paint off of screws	The SD allows the user to work with screws that have been painted over	
	Would be nice if it could punch a pilot hole	The SD can be used to create a pilot hole (!)	

Guideline	Customer Statement	Need Statement- <u>Wrong</u>	Need Statement- <u>Right</u>
<u>What Not</u> <u>How</u>	"Why don't you put protective shields around the battery contacts?"	The screwdriver battery contacts are covered by a plastic sliding door.	The screwdriver battery is protected from accidental shorting.
Specificity	"I drop my screwdriver all the time."	The screwdriver is rugged.	The screwdriver operates normally after repeated dropping.
Positive Not Negative	"It doesn't matter if it's raining, I still need to work outside on Saturdays."	The screwdriver is not disabled by the rain.	The screwdriver operates normally in the rain.
Attribute of the Product	"I'd like to charge my battery from my cigarette lighter."	An automobile cigarette lighter adapter can charge the screwdriver battery.	The screwdriver battery can be charged from an automobile cigarette lighter.
Avoid "Must" and "Should"	"I hate it when I don't know how much juice is left in the batteries of my cordless tools."	The screwdriver should provide an indication of the energy level of the battery.	The screwdriver provides an indication of the energy level of the battery.

Organizing needs

Organize needs as a hierarchy

- Print or write each need statement on a separate card or 'Post-It'
- Eliminate redundant statements
- Group cards according to the similarity of expressed needs
- For each group, choose a label
- Consider creating “supergroups” consisting of two to five groups
- Review and edit need statements

Ranking Needs

Cordless Screwdriver Survey

For each of the following cordless screwdriver features, please indicate on a scale of 1 to 5 how important the feature is to you. Please use the following scale:

1. Feature is undesirable. I would not consider a product with this feature.
2. Feature is not important, but I would not mind having it.
3. Feature would be nice to have, but is not necessary.
4. Feature is highly desirable, but I would consider a product without it.
5. Feature is critical. I would not consider a product without this feature.

_____ The screwdriver can drive screws into hardwood.

_____ The screwdriver can turn phillips, torx, socket, and hex head screws.

_____ The screwdriver can access screws at the end of deep, narrow holes.

And so forth.

The SD provides plenty of power to drive screws.

- * The SD maintains power for several hours of heavy use.
- ** The SD can drive screws into hardwood.
The SD drives sheet metal screws into metal ductwork.
- *** The SD drives screws faster than by hand.

The SD makes it easy to start a screw.

- * The SD retains the screw before it is driven.
- !* The SD can be used to create a pilot hole.

The SD works with a variety of screws.

- ** The SD can turn philips, torx, socket, and hex head screws.
- ** The SD can turn many sizes of screws.

The SD can access most screws.

- The SD can be maneuvered in tight areas.
- ** The SD can access screws at the end of deep, narrow holes.

The SD turns screws that are in poor condition.

- The SD can be used to remove grease and dirt from screws.
- The SD allows the user to work with painted screws.

The SD feels good in the user's hand.

- *** The SD is comfortable when the user pushes on it.
- *** The SD is comfortable when the user resists twisting.
 - * The SD is balanced in the user's hand.
 - ! The SD is equally easy to use in right or left hands.
 - The SD weight is just right.
 - The SD is warm to touch in cold weather.
 - The SD remains comfortable when left in the sun.

The SD is easy to control while turning screws.

- *** The user can easily push on the SD.
- *** The user can easily resist the SD twisting.
 - The SD can be locked "on."
- ! ** The SD speed can be controlled by the user while turning a screw.
 - * The SD remains aligned with the screw head without slipping.
- ** The user can easily see where the screw is.
 - * The SD does not strip screw heads.
 - * The SD is easily reversible.

The SD is easy to set-up and use.

- * The SD is easy to turn on.
- * The SD prevents inadvertent switching off.
- * The user can set the maximum torque of the SD.
- ! * The SD provides ready access to bits or accessories.
- * The SD can be attached to the user for temporary storage.

The SD power is convenient.

- * The SD is easy to recharge.
The SD can be used while recharging.
- *** The SD recharges quickly.
The SD batteries are ready to use when new.
- ! ** The user can apply torque manually to the SD to drive a screw.

The SD lasts a long time.

- ** The SD tip survives heavy use.
The SD can be hammered.
- * The SD can be dropped from a ladder without damage.

The SD is easy to store.

- * The SD fits in a toolbox easily.
- ** The SD can be charged while in storage.
The SD resists corrosion when left outside or in damp places.
- !* The SD maintains its charge after long periods of storage.
The SD maintains its charge when wet

The SD prevents damage to the work.

- * The SD prevents damage to the screw head.
The SD prevents scratching of finished surfaces.

The SD has a pleasant sound when in use.

The SD looks like a professional quality tool.

The SD is safe.

- The SD can be used on electrical devices.
- *** The SD does not cut the user's hands.

PAY ATTENTION

TEAM BONDING ACTIVITY



PLEASE TAKE NOTE

Should time permit: Use SCRAP MATERIALS, straws to cushion an egg falling

Team Finalization (Team Name etc)

Thursday Green

Lau Zhi Hsien Josiah
Oon Ewe Huat
Zhengchengsheng
Wangzhenrong
Sai-Chin-Kai
Foo Lai Wah

Thursday Blue

Ding Tianqi
Yang Qingwei
Chan Kok Siang
Duan Xu
Tan Ai Kun
Goh Zi An Galvyn

Thursday Red

Prathosh Chander
Rahul Sharan Suriya Kumar
Mehta,Manan Bimal
Sagar Pratap Singh
Cheng Zihao
Louisa Ruth Nonis

Thursday Orange

Lourduraj John Charles
Yang Xingyu
Ang Kar Kin
Syed Adeel Ahmed
Andaramanage Nisal Deelaka Jinadasa

Thursday Purple

Nguyen Thanh Tung
Lai Siew Wen
Chua Kwang Hai Andrew
Nicolas Lasch
Lee Kai Boo

Team Formation Activity

02.B Student Profile

Put your best pic here!

About me:

Write 1-2 sentences about yourself. Include your interests, focus areas, or what you enjoy working on.

Student Name :
Student 1
Student ID :
12345678

Designs I love the most:

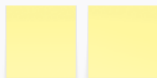
Share 1-2 designs, products, or projects that inspire you. It could be because of their creativity, functionality, or how they make you feel.

02.A Team Charter

Click the link, design your avatar with colors and quirks. Once done, copy and drop it onto the Icebreaker Board so your character becomes part of our team lineup.

What kind of team do we want to be, and what's one thing we want to do really well together?

PURPOSE: Discussion about team culture, collaboration style, and shared mindset.



Collaboration Style

Student Name :

I feel most engaged when

and I approach it by

Student Name :

I feel most engaged when

and I approach it by

Student Name :

I feel most engaged when

and I approach it by

Student Name :

I feel most engaged when

and I approach it by

Student Name :

I feel most engaged when

and I approach it by

What excites us most about this project, and what do we hope to learn along the way?

PURPOSE: Encourages the team to share their motivation and curiosity, highlights what each member values.



	Studen...	Student 1	Student 2	Student 3	Student 4	Student 5
1	CAD Modelling & Digital Design	Confident		Confident		Comfortable
2	Coding		Just Starting		Just Starting	
3	Electronics			Confident		Just Starting
4	Prototyping & Building		Comfortable			
5	Visual Communication & Graphic Design				Comfortable	
+						

Use this exercise to explore potential roles and strengths within the team.

Team Formation Activity (Icebreakers)

We want teams with different disciplines and backgrounds but similar interests (Research/Project-wise)

Team Formation

- - How much experience/interest do you have certain software CAD & Mech Design, AI Design Tools ?
- Different cultural/discipline backgrounds – At least 2
- How much experience/interest do you have in designing Fabrication/Prototyping?
- Keep group size to 5-6 students

After which when you are unable to arrange, we will assign you

Team Formation Activity (Icebreakers)

Questions:

1. How will you manage your project plan? How often will you meet as a team?
2. Who will be the team lead to keep everyone on track and move decisions along? If you're used to playing the role of a lead, we encourage you to step to the side and support one of your team members in taking on this role so that they get practice.
3. What are the roles within your team (biz, eng, design)? We highly encourage each team member to take on a role that's different from their current background.
4. How will you hold one another accountable? If a team member falls behind on their responsibilities, how will you address it?

Team Formation Activity (Icebreakers)

Questions:

5. How do you manage stress?
6. What's one of your biggest pet peeves when working in a team and how did you deal with it?
7. What do you define as success for your team with this project?
8. What are your personal goals with this project?
9. What type of technical support and training will your team need the most help with to make this a successful project?
10. What's your team name?

GET TO KNOW EACH OTHER

Team Formation Activity (Icebreakers)

We want teams with different disciplines and backgrounds but similar interests (Research/Project-wise)

Team Formation

- - How much experience/interest do you have certain software CAD & Mech Design, AI Design Tools ?
- Different cultural/discipline backgrounds – At least 2
- How much experience/interest do you have in designing Fabrication/Prototyping?
- Keep group size to 5-6 students

After which when you are unable to arrange, we will assign you

Team MIRO board

LC01 link: Monday Morning Class

[https://miro.com/app/board/uXjVHnzM9vo=/?share link id=809007457493](https://miro.com/app/board/uXjVHnzM9vo=/?share_link_id=809007457493)

LC02 Link: Monday Afternoon Class

[https://miro.com/app/board/uXjVHntA6SI=/?share link id=932679953](https://miro.com/app/board/uXjVHntA6SI=/?share_link_id=932679953)

LC03 Link: Monday Evening

[https://miro.com/app/board/uXjVHnykHmw=/?share link id=275232710964](https://miro.com/app/board/uXjVHnykHmw=/?share_link_id=275232710964)

LC05 Link: **Thursday Afternoon Class**

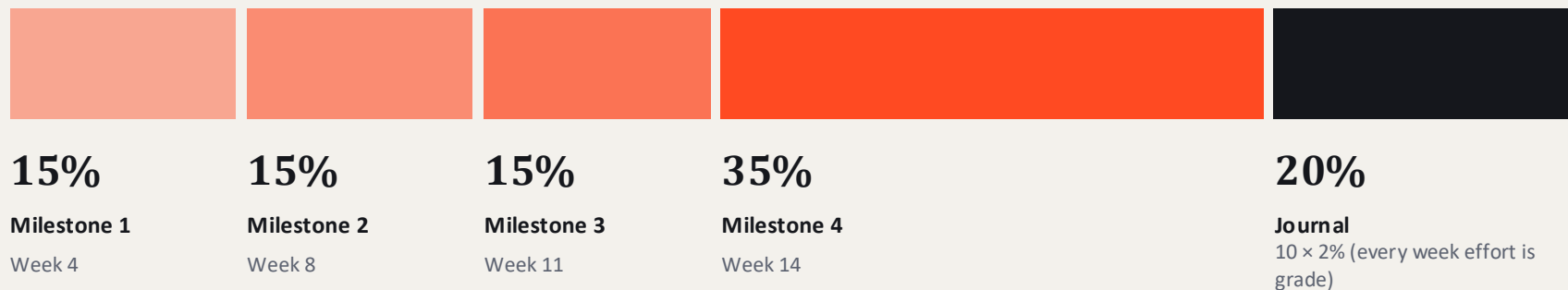
[https://miro.com/app/board/uXjVHnzA8E0=/?share link id=35289043780](https://miro.com/app/board/uXjVHnzA8E0=/?share_link_id=35289043780)

LC04 Link: Thursday Evening Class

[https://miro.com/app/board/uXjVHn3fwoc=/?share link id=30411990514](https://miro.com/app/board/uXjVHn3fwoc=/?share_link_id=30411990514)

How the 100% breaks down

There are no weekly quizzes. Each milestone is marked out of 100 and converted to its weighting.



 Milestone grades are awarded at team level, then moderated by two peer reviews (Weeks 8 and 14).

 Milestone 4 is marked by a panel of three to four faculty; your score is the average of their ratings.

 High sales alone do not guarantee a high grade. Strategy, execution, learning and evidence do.

INDICATIVE SCHEDULE

Fourteen weeks

1	14–18 Sep	Introduction, teams, AI toolchain	Journal 1	8	2–6 Nov	Milestone 2 · peer review	Presentation
2	21–25 Sep	Discover: interviews, AI research	Journal 2	9	9–13 Nov	Deliver: prototyping and CAD	Journal 7
3	28 Sep – 2 Oct	Define: scoping and benchmarking	Journal 3 · M1 due	10	16–20 Nov	DFMA, costing, unit economics	Journal 8 · M3 due
4	5–9 Oct	Milestone 1 · Venture Thread	Presentation	11	23–27 Nov	Milestone 3 · Venture Readiness	Presentation
5	12–16 Oct	Develop: ideation, generative AI	Journal 4	12	30 Nov – 4 Dec	Manufacture and launch prep	Journal 9
6	19–23 Oct	Develop: selection and price	Journal 5 · M2 due	13	7–11 Dec	Launch, sell, fulfil, iterate	Journal 10 · M4 due
7	26–30 Oct	Break week · synthesis of feedback	Journal 6	14	14–18 Dec	Milestone 4 · Exhibition & Gala	Presentation

Highlighted weeks are presentation weeks. Week 7 is a break week, and still carries a journal entry.

PUT THESE IN YOUR CALENDAR

Four dates that decide the term

5–9

OCT

Milestone 1

Discover and Define · 15%

Checkpoint 1: sell the concept

2–6

NOV

Milestone 2

Develop · 15%

Mid-term peer review

23–27

NOV

Milestone 3

Deliver and Venture Readiness
· 15%

Checkpoint 2: sell the prototype

14

DEC

Milestone 4

Exhibition, Gala and Final
Iteration · 35%

Monday. Checkpoint 3: sell you
process and 2.0 version

Week 1 runs 14–18 September 2026. Week 7 (26–30 October) is a break week. Milestone materials are due the week before each presentation.

Short break ... 😊



Strategizing Interviews/ Discover Data Collection



Methods

- Web scraping with Claude
- ...

Week 2 –Homework dicussions

Start discussing in your teams over the next 2 weeks, potential problems you will like to solve, not just you, within team, form a google doc, form your chat groups, etc. Each person to generate their HMWs

Format: How might we develop a way to
_____ (objective)
for _____ (end user/stakeholders)
in a certain _____ (situation/environment)
How to measure success? _____ -

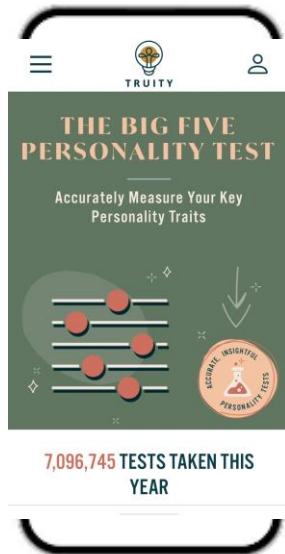


Please take these tests now and note down your result

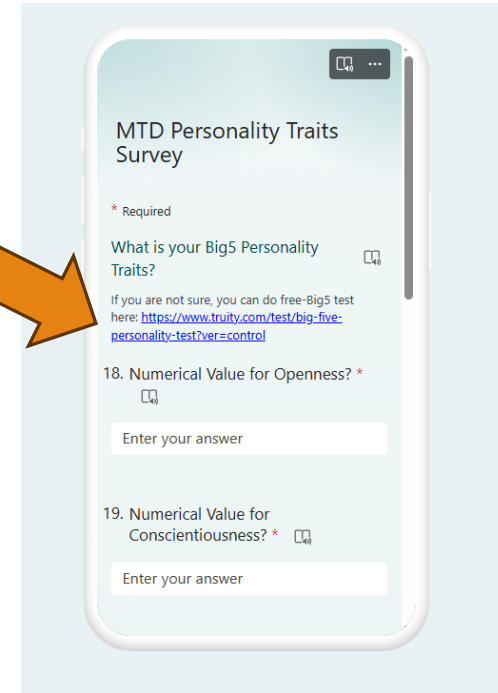
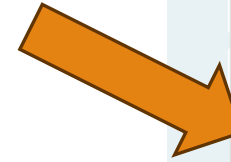


Big 5:

<https://www.truity.com/test/big-five-personality-test?ver=control>



Just scroll down and start taking the test



Please take these tests now and note down your result

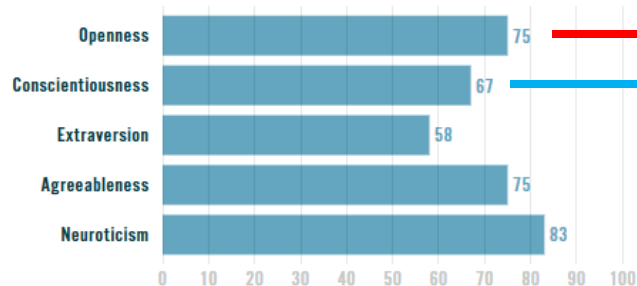


Big 5:

<https://www.truity.com/test/big-five-personality-test?ver=control>

Your Personality Trait Scores

This Big Five assessment measures your scores on five major dimensions of personality: Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism (sometimes abbreviated OCEAN).

A graphic of a smartphone screen displaying a survey titled 'MTD Personality Traits Survey'. The survey includes a required question: 'What is your Big5 Personality Traits?'. Below this, there are two numbered questions: '18. Numerical Value for Openness? *' and '19. Numerical Value for Conscientiousness? *'. Each question has a text input field labeled 'Enter your answer'. The smartphone screen is shown against a light blue background.

MTD Personality Traits Survey

* Required

What is your Big5 Personality Traits?

If you are not sure, you can do free-Big5 test here: <https://www.truity.com/test/big-five-personality-test?ver=control>

18. Numerical Value for Openness? *

Enter your answer

19. Numerical Value for Conscientiousness? *

Enter your answer

Submit your result for each of the three tests here

MTD Personality Traits Survey

Project title: Analysing the correlation between students' personality traits (MBTI, LSS, & Big5) and their design education

Principal Investigator:

Immanuel Hendra

PhD Student

Singapore University of Technology and Design

Tel: +65 84304323

By continuing the survey, I hereby acknowledge that:

1. I have read the information sheet that explains the use of my data in this research.
2. I understand its contents and agree to provide my input for the survey for the use of this research.
3. My participation in this study is completely voluntary. I can withdraw from the research at any point during filling the questionnaire by simply closing the browser window.
4. Any personal information (my ID, scores) that is collected in this study that can be associated directly to me remains confidential. My personal identifiers will be coded to protect my privacy. My personal information will be discarded when the research has been completed.
5. The research team will keep my data in a secure place where only the research team will be able access.
6. My coded data (without personal identifiers) to be used in future research. I understand that all future related research will be subject to the approval of an Institutional Review Board.

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

* Required

1. What is your student ID? (e.g: 100XXXX) *

Enter your answer

MTD Personality Traits Survey



<https://forms.office.com/r/vZPzqUVD4v>